

Multipole Optimisation of Radiative Cooling Fibres

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Abstract

Radiative cooling clothing is an emerging alternative to air-conditioning for those living or working in ever-warmer climates. Cooling properties in textiles can be achieved through the introduction of microscopic pores into otherwise ordinary polymer fibres. The air-polymer index contrast supports Mie resonances for pore diameters comparable to solar wavelengths, scattering light and shielding the fabric from solar absorption. Textiles such as polyester and nylon also emit strongly in the mid-infrared atmospheric window, exporting heat into the sky. The result is cooling performance in full sun achieved with inexpensive materials that are industrially ubiquitous.

Yet, the development of cooling fibres requires navigating an enormous design space of materials, pore positions and sizes. No first-principles tools exist for simulating and optimising dense ensembles of randomly distributed scatterers. As a result, the question of what constitutes the optimal cooling fibre geometry remains unresolved. To make this question tractable, we model fibres as non-overlapping dielectric cylinders, permitting a multipole method where Maxwell's equations reduce to a linear algebra problem: a transfer matrix mapping incident and outgoing coefficients. We have implemented this method within the automatic differentiation library *PyTorch*. Consequently, figures of merit yield pre-computed gradients, which guide an efficient search for fibre geometries that maximise cooling.

Acknowledgements

An enormous thank you to Boris Kuhlmei, who has been an exceptional supervisor and the compass to this entire project. Every Wednesday morning I would stumble into his office rambling about Bessel functions or a mysterious sign error that had consumed my week. He would somehow take it all in, untangle the chaos, and restate the problem such that everything suddenly made sense again. His ability to turn my incoherence into understanding is magic, and I am deeply grateful for his friendliness and optimism throughout the year.

Thank you also to Alex Song, who had the honour of listening to the Friday instalments of my ramblings and brought a welcomed, fresh perspective whenever Boris and I were staring at the whiteboard, waiting for the physics to reveal itself.

And to Stuart Hawkins, the resident mathematician of this project, for his friendly guidance on matrix inversion strategies and abundance of ideas, which we hope to explore in the future.

Thank you to my friends and family for their encouragement and reminders to occasionally leave my desk. Oonagh, thank you for your patience, humour, and for enduring a year of “is the code working today?” - “almost!”

Statement of Student Contribution

The initial proposal to combine the multipole method with automatic differentiation for cooling textiles was developed by Prof. Boris Kuhlmei. I rederived the multipole formalism based on Kuhlmei et al. (2002) [1], since several oblique incidence scattering matrices required for this work were not readily available in the literature. The idea to azimuthally average the scattered power was put forward by Prof. Kuhlmei and derived myself. The derivation of the forward scattering figure of merit was also completed by me, following a discussion between Prof. Kuhlmei and Dr. Alex Song in which it was agreed that this was the most relevant metric for solar reflectance. The realisation that convolutions could be introduced to improve the efficiency of the algorithm and the idea to introduce an intersection penalty function was my own. All *Python* code, including the custom *PyTorch* multipole solver, geometry generation routines, optimisation framework, and statistical analysis, was developed myself. Numerical validation of the solver was greatly aided by Prof. Stuart Hawkins’ Miesolver and Prof. Kuhlmei’s Winfield simulation tools.

Statement of Originality

*I certify that this thesis contains work carried out by myself
except where otherwise acknowledged.*

Signed Date

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1 Introduction

The management of thermal energy remains a central concern, quietly governing comfort, efficiency, and the uneasy truce maintained with the entropy around us. The standard interventions: fans, compressors, refrigeration merely shift heat elsewhere, guaranteeing its eventual return as local or global warming. As incomes and global temperatures have risen, the adoption of air-conditioning and consumption of fossil fuels to power these systems has swelled, particularly in countries with hotter climates. Simultaneously, the atmospheric burden of hydrofluorocarbons rises by 0.8 Gt CO₂-equivalent per annum [2], driven by leakage from air-conditioning units and insulation foams. As a consequence, in the period from 1971 to 2020, the Earth has accumulated an estimated 381 ZJ of heat energy, with the planetary heating rate reaching 1.06 Wm⁻² in 2018 [3]. Ultimately, the only permanent route for heat removal is into space.

Space, as it turns out, functions as a near-ideal heat sink. The immense temperature gradient between the Earth's surface (300 K) and the cosmic abyss (3 K) drives radiative heat transfer. The theoretical upper bound on radiative cooling is determined by the Stefan-Boltzmann law, giving $q_{max} = \sigma(300^4 - 3^4) = 459 \text{ Wm}^{-2}$ of potential heat rejection. In practice, the atmosphere and the sun throttle cooling: intense solar radiation continually introduces additional heat, while water vapour, aerosols and CO₂ absorb and re-emit most infrared light, preventing its escape into space. Nevertheless, a spectral corridor remains: measured in 1928 by G. C. SIMPSON [4] an 8 - 13 μm window permits thermal emission through the atmosphere into deep space. This mechanism underpins not only the planet's cooling but also the ability of individual surfaces to radiatively shed heat. For an object to achieve net cooling during the daytime it must emit efficiently within this window while reflecting or scattering most incoming solar radiation. Development of such selective emitters has surged, giving rise to the diverse and expanding field of passive radiative cooling (PRC). Within the last few years alone startups such as *SkyCool* have implemented passive cooling metal-dielectric panels atop shopping centres saving refrigeration power consumption by 20,000 kWh per year [5]. In agriculture, biodegradable radiative cooling films have been developed to reduce soil temperature and moisture evaporation [6]. PRC structures can also be used to condense water-vapour to meet drinking-water scarcity, an ideal emitter achieving 13 g m⁻²hr⁻¹ [7]. This suggests a future where cooling no longer demands the whirring of compressors and fans, as excess heat vanishes silently, radiated into the sky.

While PRC panels and films are now commercially viable for buildings, billions of farmers, construction workers, and outdoor labourers rely on clothing alone to withstand extreme heat. Each year, heatwaves claim around five million lives worldwide [8], a toll expected to rise sharply as the planet warms. Yet people cannot be clad in metal-dielectric panels. Textiles must breathe, flex, endure wear, and resemble clothing rather than laboratory equipment. Achieving comparable cooling performance in fabrics is therefore far more challenging. Their architecture spans from nanometre pores to macroscopic weave patterns. Although the theoretical limits of radiative cooling are well established, identifying an optimal fibre geometry for a given material requires millions of electromagnetic simulations, a task beyond the reach of established tools. Because of this computational bottleneck, fabrication often proceeds by trial and error. This thesis overcomes this

limitation, introducing a rapid simulation and optimisation framework for porous polymer fibres that can provide quantitative guidance for the next generation of life saving cooling clothing.

The thesis is structured as follows. Chapter 2 reviews the historical development and physical principles of radiative cooling. Chapter 3 introduces the theoretical foundations of the multipole method and its application to fibre solar scattering. Chapter 4 details the numerical implementation of the method and an accompanying optimisation algorithm. Chapter 5 presents the results and analysis of porous fibre solar scattering. Chapter 6 concludes the thesis and outlines potential extensions and applications of this work.

2 Literature Review

2.1 The History of Radiative Cooling

PRC was first conceptualised in 1804 by the Count of Rumford [9] upon the realisation that a cold blackened metal surface would cool a nearby object more effectively than a bare metal surface. He proposed a theory describing the transfer of cooling rays.

“frigorific rays arrive continually at the surface of the earth from every part of the heavens; and it is no doubt by the action of these rays that our planet is continually cooled... ...withstanding the immense quantities of heat that are generated at its surface by the continual action of the solar rays. The action of these frigorific nocturnal rays will likewise justify the inhabitants of hot climates, who, in order to be more cool during their hours of rest, remove their beds in summer to the tops of their houses”

In 1928, Arago reported on nocturnal radiative cooling in fibrous materials, noting that small amounts of grass, cotton or duck feathers placed under a clear night sky reached temperatures 6 – 8° C below ambient temperature [10]. During the day however, radiative cooling is negligible compared to radiative heating from the sun. Consequently, interest in passive radiative cooling lay dormant until the oil crisis of the 1970s which sparked a renewed interest in energy research. In 1975, S. Catalanotti et al. [11] theoretically demonstrated that a surface with high emissivity in the 8 - 13 μm window and strong solar reflectance could cool below ambient temperature under sunlight. What followed was decades of attempts to engineer such surfaces experimentally, tuning the geometry and material properties to achieve the required spectral selectivity. By 1978 Harrison and Walton [12] identified that commercial titanium oxide paint functions as a selective emitter within the atmospheric window, though it failed to sustain sub-ambient temperatures under direct sunlight. It became clear that achieving daytime radiative cooling required far finer spectral control achieved through a combination of accurate simulation and specialised fabrication. By 1995, radiative transfer simulations were employed to optimise ZnS pigmented polyethylene foils [13]. These structures managed to cool below ambient temperature for 19 hours of the day, but true all day cooling remained out of reach. It was not until 2013 that Rigorous Coupled Wave Analysis (RCWA) enabled the design of a multilayer metal dielectric stack capable of continuous sub ambient cooling [14]. Within a year, the design was experimentally realised and sustained 4.9°C below ambient under full sunlight for a full day, the first continuous demonstration of daytime radiative cooling. [15].

This achievement highlighted that sophisticated simulation and optimisation were essential to advance radiative cooling design. As the field of daytime PRC has matured, similar cooling efficiencies have been realised across diverse material systems, each exploiting different mechanisms of spectral control. To understand these advances, it is necessary to examine the underlying theory governing radiative heat exchange: how emission, absorption, and reflection balance to determine the net cooling power and the strategies by which emissivity can be tuned.

2.2 Physics of Radiative Cooling

Any object at finite temperature emits and absorbs thermal radiation through electromagnetic radiation. Placed under a clear sky, an emitter will radiate some amount of energy P_{emi} , and absorb energy from atmospheric emission P_{atm} , solar radiation P_{sun} and non-radiative transfer P_n [16] depicted in Fig. 2.1.

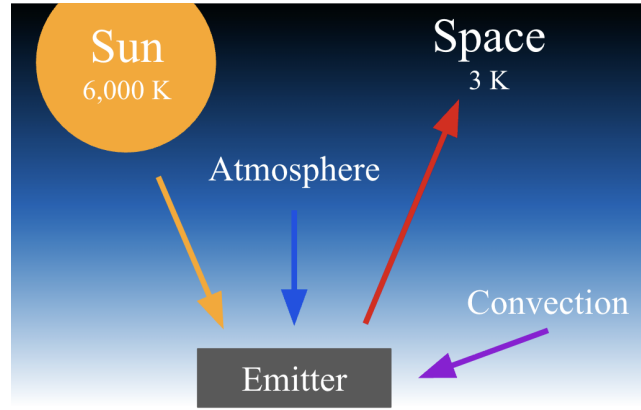


Figure 2.1: Schematic of radiative heat exchange between an emitter and its surroundings.

The balance between the incoming and outgoing power limits the net cooling of the structure. The net cooling power is defined as

$$P_{cool} = P_{emi} - P_{atm} - P_{sun} - P_n. \quad (2.2.1)$$

If we wish to achieve $P_{cool} > 0$ the power emitted by the structure must be greater than the power absorbed. In wearable cooling, non-radiative transfer of bodily heat is unavoidable; however, the radiated power can be controlled. By tailoring the emissivity ϵ , the absorbed radiation P_{atm} , P_{sun} , can be reduced and the emitted radiation P_{emi} enhanced to become the dominant term in the equation. The power emitted P_{emi} by a passive cooling structure is expressed as

$$P_{emi} = A \int d\Omega \cos \theta \int_0^\infty d\lambda I_{BB}(T, \lambda) \epsilon(T, \lambda, \Omega). \quad (2.2.2)$$

Here A denotes the area of the cooling structure, Ω the solid angle, λ the wavelength of emitted radiation, and θ the emission angle measured from the surface normal. The emitted power is dependent on temperature, governing its black body radiation I_{BB} and most critically its emissivity profile $\epsilon(T, \lambda, \Omega)$, which may be manipulated by carefully tuning the materials and shape of the structure.

In a reciprocal medium, Kirchhoff's law of thermal radiation holds for every wavelength and angle, requiring $\epsilon(T, \lambda, \Omega) = \alpha(T, \lambda, \Omega)$. Therefore, a surface that emits efficiently at a given wavelength and direction will also absorb efficiently under illumination of the same conditions. To maximise P_{emi} , one would want a black body, for which $\epsilon = \alpha = 1$. However, such broadband absorption would be disastrous for cooling, as strong thermal emission implies equally strong absorption of incoming solar and atmospheric radiation, both of which greatly exceed the emitters thermal output. It is therefore essential to tailor emissivity to the spectral shapes of the source radiation. Fortunately, the solar and thermal infrared spectra are largely distinct (Fig. 2.2).

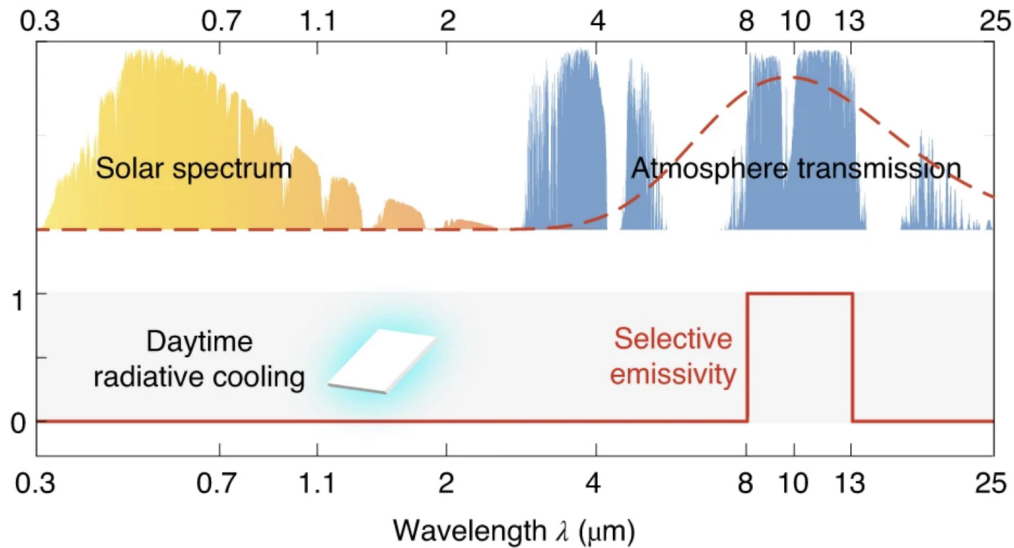


Figure 2.2: Schematic of radiative spectra adapted from Fan & Li [16]. Solar spectrum (yellow) and the atmospheric transmission spectrum (blue). Black body radiation spectra at 300K (red dashed curve). Selective emitter spectral emissivity (red solid curve).

At 6000 K the sun's black body radiation peaks at $\lambda \approx 0.5\mu\text{m}$ with the majority of power radiated over a spectral region from 0.3 - 2.5 μm . Conversely, the atmosphere and terrestrial surfaces, near 300 K, emit predominantly in the mid-infrared, peaking around 10 μm . However, much of this radiation is absorbed and re-emitted by water vapour and CO_2 , leaving only a 8 – 13 μm atmospheric window through which heat can escape directly to space. Emission outside this window is trapped and re-radiated merely heating the air around the emitter instead of dissipating energy to space. A passive cooling structure must therefore selectively emit in this window and reflect all other radiation, as depicted by the solid red curve in Fig. 2.2. The next section explores the various material and structural strategies used to achieve selective emissivity, with particular attention to approaches applicable to textiles.

2.3 Engineering Emissivity

The human body naturally emits most of its thermal radiation within the 8 – 13 μm atmospheric window. Textiles can therefore assist radiative heat loss in two distinct ways: they may be infrared transparent, allowing bodily emission to pass through freely, or infrared emissive, acting as the radiator themselves. For emissive designs, heat retention between the skin and textile must be minimized

through a skin-tight fit. In either case, the structure must also minimise solar absorption to prevent external heating. Designs that meet these criteria generally fall into three categories: porous structures, embedded particles, and multilayered composites.

2.3.1 Porous Structures

The simplest way to achieve strong solar reflection, and the method that evolution developed, is to introduce many small pockets of air into the structure. For example, the cocoon of the Madagascan comet moth contains a dense network of elongated air voids running along its length (Fig. 2.3). These voids are roughly a few hundred nanometres to a micron across, comparable to visible and near-infrared wavelengths. The refractive index contrast between the silk matrix $n = 1.5$ and the trapped air $n = 1$ produces scattering of incoming light. With each scattering event, a fraction of light is redirected outward while the rest continues deeper into the fibre. Photons that leave the material do not return, whereas those that travel onward encounter further scatterers. Over many such events, the cumulative effect is a high probability of back reflection of solar radiation (Fig. 2.3). This gives the cocoon a characteristic silvery white appearance and keeps the pupa cool under direct sunlight. The morphology of the moth cocoon fibres bear striking resemblance to micro-structured optical fibres. In later sections we will apply the same formalism originally developed for electromagnetic scattering of optical fibres to simulate cooling textiles fibres.

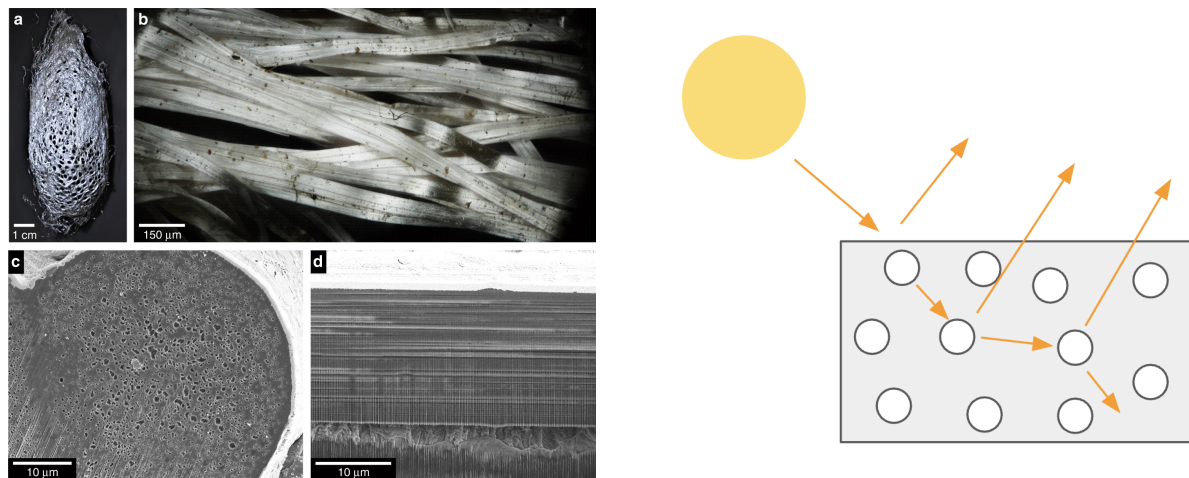


Figure 2.3: Left: a) Photograph of a comet moth cocoon. b) Dark-field optical microscopy image showing overlapping cocoon fibres. c) Cocoon silk fibre transverse cross section. d) Longitudinal cross section of a silk fiber, adapted from Shi *et al.* [17] Right: Schematic illustration of multiple scattering within a porous structure.

The multiple scattering effect, seen in biological systems, has been realised in engineered nano-porous polyethylene (nanoPE) [18]. Ordinary polyethylene, such as food wrap, is inherently transparent to visible light and infrared radiation, allowing thermal emission from the body to escape. By introducing a hierarchy of air pores spanning nanometre to micrometre diameters, each pore scatters a different portion of the solar spectrum, producing broadband Mie scattering that renders nanoPE opaque. Unfortunately, wearing a sheet of nanoPE film would be about as pleasant as wrapping oneself in cling wrap. To make the material breathable nanoPE has been spun into fibres and transformed into a fabric resembling cotton [19, 20].

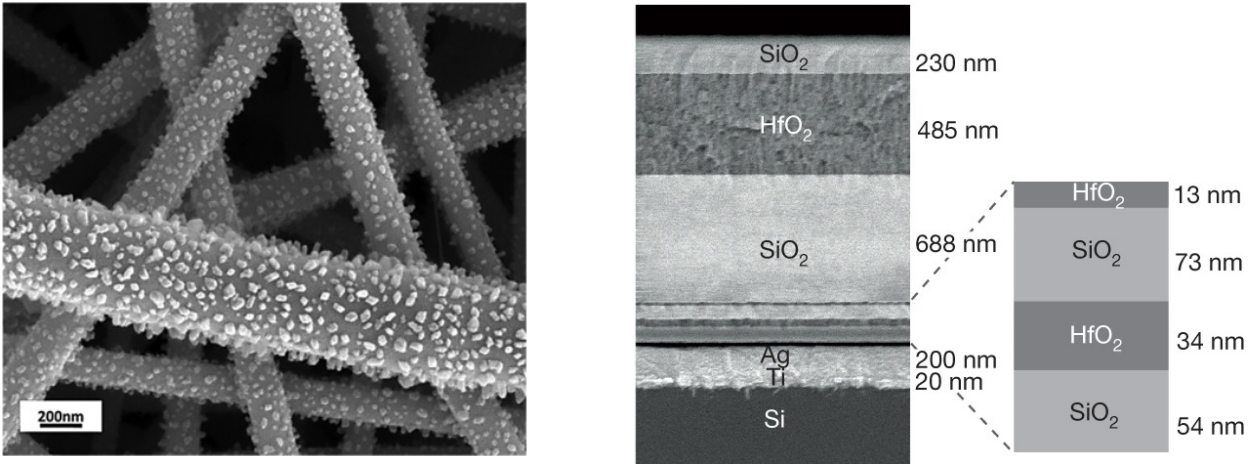


Figure 2.4: *Left:* SEM image of Cu nanoparticles deposited on surface of fibre from Ge *et al.* [28]. *Right:* SEM cross section of a multi layered dielectric-metal stack, adapted from Raman *et al.* [15].

2.3.2 Inclusion of Embedded Particles

Similar in mechanism to nano-porous textiles, the incorporation of nanoparticles (NPs) with a similar size to incident radiation into a fabric results in bulk reflection through collective Mie Scattering (Fig. 2.4). A range of materials may be used for NPs such as TiO₂ and Al₂O₃ [21]. NPs have even been attached to ubiquitous natural fibres such as silk [22]. NPs are not just suitable for solar reflection: many materials such as NaH₂PO₂ [23], SiO₂ and Al₂O₃ [24] can amplify infrared emission through coupling with surface phonon polaritons, further assisting radiative cooling.

2.3.3 Multilayer Composites

The first selective emitter to achieve all-day radiative cooling was a seven layer stack of dielectric oxides on a silver mirror [15] (Fig. 2.4). Multilayered textiles share a similar structure: a metallic back layer that is highly reflective across the solar band such as Ag [25] or Al [26]. Above this layer highly infrared emissive polymers such as polyformaldehyde (POM) and polytetrafluoroethylene (PTFE) [26] boost the radiative emission of the structure in the atmospheric window. There have also been efforts to augment cotton with a top layer of nano-porous film to boost reflectivity while maintaining the comfort of cotton [27].

High performance radiative cooling textiles often combine porous, nanoparticle, and multilayer designs. Yet greater structural complexity brings costly and impractical fabrication. Textile production demands processes that can scale to thousands of kilometres of fibre while avoiding unstable or skin staining materials such as metal coatings. Porous polymer fibres therefore offer one of the most practical paths to commercial cooling clothing. This raises a central question: how far can radiative cooling be pushed using polymers and internal air structure alone? To address this, we examine the simulation tools currently available and the computational barriers that halt the discovery of optimal structures.

2.4 Simulation of Radiative Cooling Structures

The progress of radiative cooling design is ultimately limited by the sophistication of electromagnetic simulations. Early breakthroughs were driven by Rigorous Coupled Wave Analysis (RCWA), ideally suited to modelling periodic planar multi-layers. When paired with optimisation strategies such as the Needle algorithm [29], RCWA facilitated the design of the first experimentally realised daytime radiative cooling structures [14, 15].

Recently, development of passive cooling textiles has extensively used RCWA to predict mid-infrared transmissivity [19, 22, 30, 31]. Yet textiles are far from a periodic planar material. Textile fibres simulated in RCWA are often approximated as square pillars arranged in a periodic lattice, an assumption that neglects curvature and irregular fibre packing. For more accurate geometric modelling, general purpose techniques such as FDTD [32, 33] and FEM [34–36] have been adopted. However, solvers capable of handling arbitrary geometries require spatial meshing where boundary conditions equations must be computed at each edge. These models suffer from large amounts of running memory, especially in 3D and at short wavelengths which require smaller grids and step sizes to maintain accuracy. These shortcomings typically lead to simplifications such as the omission of the ultraviolet band as noted in a recent paper by Niu et al. (2024) [32]. The commercial availability and generality of FEM and FDTD software makes these approaches widely adopted, albeit faster computational methods are highly desirable.

Optimising the geometry of porous fibres is computationally exhaustive. The parameter space is extremely high dimensional (each air hole contributing a position and radius) even a modest fibre containing 50 holes represents a 150+ dimensional optimisation problem. Each candidate geometry must be simulated over hundreds of wavelengths spanning the solar and thermal infrared bands, and across all incidence angles to capture oblique scattering. A single optimisation run explores hundreds of geometries and therefore requires tens of thousands of electromagnetic simulations. Optimisation is therefore only possible if each forward solve completes in well under a second, ideally in the 10 – 100 ms range, with easily accessible gradients for rapid updates.

Because of this computational cost, rigorous electromagnetic treatments are rarely applied to the solar scattering of textile fibres. Instead, many studies use mean field models based on the multipole expansion of a single cylinder [20], sometimes combined with Monte Carlo ray tracing [21, 37]. These methods capture far field scattering reasonably well for sparse fibre assemblies but neglect the multiple scattering and near field coupling that arise in dense textile structures.

The generalised multipole method originally developed for micro structured optical fibres by White and Kuhlmeier et al. [1, 38] offers a path beyond these limits. By exploiting the symmetry of circular pores, it expresses boundary conditions analytically, eliminating spatial meshing and directly solving for the fields in the surrounding medium. This facilitates multiple scattering simulation at a fraction of the cost of FEM or FDTD.

This thesis implements, for the first time, a multipole solver that computes derivatives in parallel. This is made possible through automatic differentiation, which records each numerical operation in a graph tracking the influence of geometric parameters on scattering. What was a slow, trial and error simulation problem is now a direct, gradient driven route to the best possible cooling fibre.

2.4.1 Thesis Outline

Radiative cooling in polymer fibres depends on how their internal geometry controls the scattering and emission of light. The best fibre we can design is ultimately limited by the speed of our numerical search. Therefore, the goal of this thesis is to develop a gradient aware multipole method and optimisation routine to find optimal cooling fibres. The theoretical formulation is presented in Chapter 3, followed by its implementation within the automatic differentiation library *PyTorch* in Chapter 4. In Chapter 5 we leverage this model to uncover the geometric motifs that govern broadband scattering in porous fibres to infer design principles for the next generation of radiative cooling textiles.

3 Theory

The Multipole Method forms the analytical foundation and principal workhorse of this study. It provides a highly accurate description of how structured materials scatter electromagnetic radiation and is particularly well suited to systems composed of cylindrical scatterers. The theoretical foundation of multipole scattering originates with Lord Rayleigh in the late 19th century, who first developed the approach to investigate wave propagation through small cylinders or spheres [39]. Many extensions to his approach have been developed over the last century. The multipole scattering of a random set of parallel cylinders was introduced by Felbacq in 1994 [40] and the theory was generalised to oblique incidence and fibre jackets for the analysis of optical fibres by Kuhlmeiy and White [38], [1] in 2002. This section introduces the Multipole Method and builds on the formalism of White and Kuhlmeiy through additional scattering matrices not included in previous work, as well as newly derived quantities relevant to radiative cooling.

3.1 The Multipole Method

We begin by representing the electromagnetic fields in a Fourier Bessel series, decomposes the fields into angular harmonics that align with the cylindrical geometry. The scattering equations are then derived at a single cylindrical boundary to establish a link between incoming and scattered fields. The single cylinder formulation is then extended using Graf’s addition theorem, which accounts for translations of fields between neighbouring cylinders, facilitating the assembly of a global scattering matrix describing all multiple scattering events. In this work, new quantities relevant to passive cooling are derived from the global scattering matrix. Crucially, we introduce an averaged measure of fibre scattering in the forward direction, reducing the number of simulations to characterize a fibre by two orders of magnitude.

3.1.1 Fibre Geometry

Porous polymer fibres are practical to fabricate and achieve high optical contrast without surface coatings or embedded particles. The longitudinally invariant nature of extrusion fabricated porous fibres makes them directly compatible with the cylindrical symmetry assumed in the multipole method. To maintain structural regularity in extrusion, air pores must remain spatially isolated, forming non-overlapping cylindrical geometries. This is ideally suited to the multipole method, which

relies on the fundamental assumption that inclusions are non-overlapping (fulfilling the hypotheses of Graf's Theorem). In principle, the multipole framework can be extended to cylinders with arbitrary cross-sectional shapes. However, circular cylinders have analytic basis functions, making the formulation significantly more efficient. The transverse (r, θ) cross section of such a fibre is pictured in Fig. 3.1, the z -axis points along the invariant length of the fibre. The fibre is perforated with a finite number of cylindrical holes that extend along the z -axis. The fibre is composed of a dielectric with $n_f \approx 1.5$, representative of common textile polymers such as polyester or nylon, holes and regions outside of the fibre are modelled as air-filled regions $n_h = 1$.

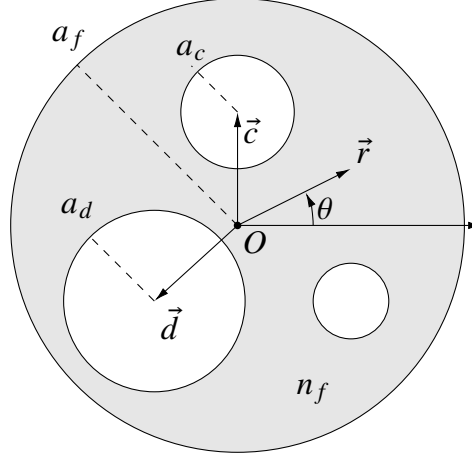


Figure 3.1: Porous fibre geometry in a polar coordinate system with notation used in the formulation of the multipole method. The fibre is centred at the origin, non-overlapping holes may be placed in any position inside the fibre $\vec{c}, \vec{d}$ with varying radii a_c, a_d .

3.1.2 Single Cylinder Scattering

The electromagnetic response of porous fibres may be modelled by Maxwell's equations. Within each constituent region of the fibre (air or polymer), the medium is assumed to be a linear, isotropic and homogenous dielectric. Under these assumptions, Maxwell's curl equations combine to a set of Helmholtz wave equations, describing the propagation of either electric and magnetic fields $\vec{\mathcal{V}} \in \{\vec{\mathcal{E}}, \vec{\mathcal{K}}\}$ within homogenous regions of the fibre. Here $\vec{\mathcal{K}} = Z\vec{\mathcal{H}}$ is the rescaled magnetic field, with similar order of magnitude as $\vec{\mathcal{E}}$ where $Z = \sqrt{\frac{\mu_0}{\epsilon_0}}$ is the vacuum impedance. The Helmholtz Equation has the form

$$\left(\nabla^2 - \frac{n^2}{c^2} \frac{\partial^2}{\partial t^2} \right) \vec{\mathcal{V}}(r, \theta, z, t) = 0. \quad (3.1.1)$$

Here n denotes the refractive index of the particular region and ∇^2 represents the Laplacian. Around each hole, the electromagnetic fields $\vec{\mathcal{V}}$ are expressed in a local cylindrical coordinate system, which simplifies the enforcement of boundary conditions at their interfaces. Although solving the Helmholtz equation in cylindrical coordinates is mathematically more involved, the additional effort is rewarded by a formulation that is both physically transparent and computationally efficient. For a single circular scatterer, each incoming cylindrical wave excites only the corresponding outgoing cylindrical wave. By contrast, in a Cartesian (plane wave) basis, a single incident plane wave would excite an infinite spectrum of outgoing plane waves at a circular boundary.

To prepare the fields for the application of cylindrical boundary conditions, it is convenient to express them in a form that reflects the symmetry of the problem. In principle, one may Fourier decompose the electromagnetic fields into plane waves over all spatial and temporal dimensions. However, since the structure is invariant in time and along the z -axis, it will become useful to separate only the temporal and longitudinal dependence, introducing the angular frequency $\omega = k_0 c$ and longitudinal wavenumber k_z . Each Fourier component satisfies the Helmholtz equation independently

$$\left(\nabla^2 - \frac{n^2}{c^2} \frac{\partial^2}{\partial t^2} \right) \vec{V}(r, \theta) e^{i(k_z z - \omega t)} = 0. \quad (3.1.2)$$

Leaving the remaining analysis confined to the transverse (r, θ) plane where boundary conditions will be applied. Evaluating the derivatives and factoring out the time and z dependence yields a 2D Helmholtz equation

$$\left(\nabla_{\perp}^2 + k_{\perp}^2 \right) \vec{V}(r, \theta) = 0. \quad (3.1.3)$$

Where $\nabla_{\perp}^2$ is the 2D transverse Laplacian, and $k_{\perp} = \sqrt{n^2 k_0^2 - k_z^2}$. In a uniform dielectric, the transverse components $E_{\theta}, E_r, K_{\theta}, K_r$ of the electromagnetic field may be formulated in terms of the longitudinal components E_z, K_z from Maxwell's curl equations [41]:

$$\begin{aligned} E_r &= \frac{i}{k_{\perp}^2} \left[k_z \frac{\partial E_z}{\partial r} + \frac{k_0 \mu_r}{r} \frac{\partial K_z}{\partial \theta} \right] & K_r &= \frac{i}{k_{\perp}^2} \left[k_z \frac{\partial K_z}{\partial r} - \frac{k_0 \epsilon_r}{r} \frac{\partial E_z}{\partial \theta} \right] \\ E_{\theta} &= \frac{i}{k_{\perp}^2} \left[\frac{k_z}{r} \frac{\partial E_z}{\partial \theta} - k_0 \mu_r \frac{\partial K_z}{\partial r} \right] & K_{\theta} &= \frac{i}{k_{\perp}^2} \left[\frac{k_z}{r} \frac{\partial K_z}{\partial \theta} + k_0 \epsilon_r \frac{\partial E_z}{\partial r} \right]. \end{aligned} \quad (3.1.4)$$

Therefore, solving the scalar Helmholtz equations for the longitudinal components $V \in \{E_z, K_z\}$ is sufficient to determine the complete electromagnetic field.

In the polar coordinate system, the azimuthal coordinate θ is periodic in 2π , consequently the electromagnetic fields must also be periodic to remain single valued $V(r, \theta) = V(r, \theta + 2\pi)$. Therefore we may expand the field as a Fourier series:

$$V(r, \theta) = \sum_{n \in \mathbb{Z}} f_n(r) e^{in\theta}. \quad (3.1.5)$$

Each integer angular order n in the Fourier–Bessel expansion represents a distinct angular harmonic of the electromagnetic field, where $n = 0, 1, 2, \dots$ corresponds to monopoles, dipoles and quadrupoles respectively. This correspondence forms the physical basis of the multipole method. In numerical implementations, higher orders make a negligible contribution to the total field and therefore the expansion can be truncated to n_{max} .

Substituting the Fourier series expansion into Eq. (3.1.3) yields

$$\left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + k_{\perp}^2 \right) \sum_{n \in \mathbb{Z}} f_n e^{in\theta} = 0. \quad (3.1.6)$$

Each angular harmonic satisfies the equation independently allowing Fourier decomposition. Computing

the azimuthal derivative and applying a change of variables $u = k_{\perp}r$ leads to the following equation:

$$u^2 \frac{\partial^2 f_n(u)}{\partial u^2} + u \frac{\partial f_n(u)}{\partial u} + (u^2 - n^2) f_n(u) = 0. \quad (3.1.7)$$

Eq. (3.1.7) is the Bessel differential equation. The solutions $f_n(r)$ to this equation are linear combinations of Bessel functions of the first $J_n(k_{\perp}r)$ and second kind $Y_n(k_{\perp}r)$ [42]. To interpret these functions physically, it is instructive to examine their asymptotic forms for large r [43]:

$$J_n(k_{\perp}r) \xrightarrow{r \rightarrow \infty} \sqrt{\frac{2}{\pi k_{\perp}r}} \cos\left(k_{\perp}r - \frac{n\pi}{2} - \frac{\pi}{4}\right) \quad (3.1.8)$$

$$Y_n(k_{\perp}r) \xrightarrow{r \rightarrow \infty} \sqrt{\frac{2}{\pi k_{\perp}r}} \sin\left(k_{\perp}r - \frac{n\pi}{2} - \frac{\pi}{4}\right). \quad (3.1.9)$$

The Sommerfeld radiation condition requires the scattered field to behave as an outward propagating wave whose amplitude decays with distance. Asymptotically, the Bessel functions represent standing waves and therefore do not satisfy the condition individually. This motivates the introduction of the Hankel functions of the first $H_n^{(1)}(k_{\perp}r) = J_n(k_{\perp}r) + iY_n(k_{\perp}r)$ and second kind $H_n^{(2)}(k_{\perp}r) = J_n(k_{\perp}r) - iY_n(k_{\perp}r)$ as a more physically appropriate basis, asymptotically representing purely outgoing $H_n^{(1)}$ and incoming $H_n^{(2)}$ electromagnetic waves with a singularity at their origin $r = 0$:

$$H_n^{(1)}(k_{\perp}r) \xrightarrow{r \rightarrow \infty} \sqrt{\frac{2}{\pi k_{\perp}r}} e^{i(k_{\perp}r - \frac{n\pi}{2} - \frac{\pi}{4})} \quad (3.1.10)$$

$$H_n^{(2)}(k_{\perp}r) \xrightarrow{r \rightarrow \infty} \sqrt{\frac{2}{\pi k_{\perp}r}} e^{-i(k_{\perp}r - \frac{n\pi}{2} - \frac{\pi}{4})}. \quad (3.1.11)$$

For the multipole formulation, the particular basis $\{J_n, H_n^{(1)}\}$ is preferable numerically: solar illumination originates from an effectively infinite distance and is therefore best represented locally by the regular Bessel functions rather than incoming Hankel functions $H_n^{(2)}$ centred at exceedingly large distances. Moreover, Graf's addition theorem expresses translated fields in terms of Bessel functions of the first kind, making the $\{J_n, H_n^{(1)}\}$ basis convenient for coordinate transformations required in scattering between many cylindrical inclusions. Substituting a linear combination of these basis functions for $f_n(r)$ into the longitudinal field equations Eq. 3.1.5 yields

$$V(r, \theta) = \sum_{n \in \mathbb{Z}} [A_n^{(V)} J_n(k_{\perp}r) + B_n^{(V)} H_n^{(1)}(k_{\perp}r)] e^{in\theta} \quad (3.1.12)$$

Here, the J_n functions represent the regular incident field and the $H_n^{(1)}$ functions represent the scattered outgoing component of the electromagnetic field. The coefficients A_n and B_n describe the amplitude of the incoming and outgoing field respectively for each angular order n . In the typical scattering problem, a known incoming wave induces a scattered wave outside the boundary, as well as some transmitted field inside the cylinder. Operating in the linear regime implies a linear mapping between the incoming coefficients A_n and the outgoing coefficients B_n . The relationship between these can be derived by solving for the boundary conditions.

We denote the fields interior V_i and outside V_o the boundary. The fields are constrained by the electromagnetic continuity conditions at the boundary radius a :

$$E_o(\vec{r})_{\parallel} - E_i(\vec{r})_{\parallel} \Big|_{r=a} = 0 \quad \vec{K}_o(r)_{\parallel} - \vec{K}_i(r)_{\parallel} \Big|_{r=a} = 0. \quad (3.1.13)$$

At the boundary, reflection and transmission of the interior and exterior fields occurs. The tangential fields include E_z, K_z directly, but also E_{θ}, K_{θ} which through Eq. 3.1.4 introduce cross coupling between E_z and K_z . Since the fields share a common angular dependence $e^{in\theta}$, boundary condition equations decompose by order n reducing to a linear system of equations between radial function coefficients $A_n^E, A_n^K, B_n^K, B_n^E$. Solving the system expresses the unknown coefficients in terms of known coefficients. For example, in the absence of field sources inside the cylinder, the external reflection from a known incoming electric field $A_{n(o)}^E$ to the desired outgoing field coefficients can be written as

$$B_{n(o)}^K = R_{n(o)}^{KE} A_{n(o)}^E \quad B_{n(o)}^E = R_{n(o)}^{EE} A_{n(o)}^E, \quad (3.1.14)$$

the scattering coefficients $R_{n(o)}^{KE}, R_{n(o)}^{EE}$ depend on the boundary refractive indices n_o, n_i , radius a , frequency ω and wavenumber k_z . Similar scattering coefficients describe the transmission of an internal outgoing field $B_{n(i)}^V$ escaping from inside the boundary $T_{n(o)}^{VV}$, from outside to inside $T_{n(i)}^{VV}$ and internal reflections on the inner surface of the boundary $R_{n(i)}^{VV}$ (see Fig. 3.2). Some, but not all of the scattering coefficients could be readily found in the literature; the remainder were re-derived for this work. This derivation is provided in Appendix A, and is an exhilarating read.

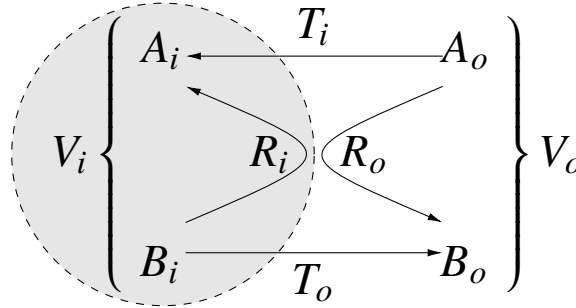


Figure 3.2: Schematic depicting the scattering matrices which relate the Bessel J_n and Hankel $H_n^{(1)}$ coefficients of the electromagnetic fields inside V_i and outside V_o at a cylindrical boundary.

To streamline later calculations and computation, the scattering coefficients are assembled into matrix form. Denoting $\vec{A}_i^{(E)} = \begin{bmatrix} A_i^{(E)} \\ A_{n(i)}^{(E)} \end{bmatrix}$ and $R_i^{(EE)} = \text{diag}(R_{n(i)}^{(EE)})$ with similar definitions for other coefficients,

$$\begin{bmatrix} \vec{A}_i^{(E)} \\ \vec{A}_i^{(K)} \end{bmatrix} = \begin{bmatrix} R_i^{(EE)} & R_i^{(EK)} \\ R_i^{(KE)} & R_i^{(KK)} \end{bmatrix} \begin{bmatrix} \vec{B}_i^{(E)} \\ \vec{B}_i^{(K)} \end{bmatrix} + \begin{bmatrix} T_i^{(EE)} & T_i^{(EK)} \\ T_i^{(KE)} & T_i^{(KK)} \end{bmatrix} \begin{bmatrix} \vec{A}_o^{(E)} \\ \vec{A}_o^{(K)} \end{bmatrix} \quad (3.1.15)$$

$$\begin{bmatrix} \vec{B}_o^{(E)} \\ \vec{B}_o^{(K)} \end{bmatrix} = \begin{bmatrix} T_o^{(EE)} & T_o^{(EK)} \\ T_o^{(KE)} & T_o^{(KK)} \end{bmatrix} \begin{bmatrix} \vec{B}_i^{(E)} \\ \vec{B}_i^{(K)} \end{bmatrix} + \begin{bmatrix} R_o^{(EE)} & R_o^{(EK)} \\ R_o^{(KE)} & R_o^{(KK)} \end{bmatrix} \begin{bmatrix} \vec{A}_o^{(E)} \\ \vec{A}_o^{(K)} \end{bmatrix}, \quad (3.1.16)$$

or equivalently

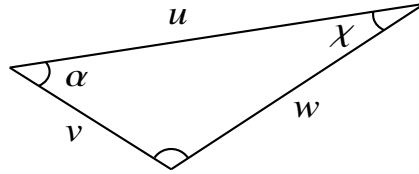
$$\mathbf{A}_i = R_i \mathbf{B}_i + T_i \mathbf{A}_o \quad (3.1.17)$$

$$\mathbf{B}_o = T_o \mathbf{B}_i + R_o \mathbf{A}_o . \quad (3.1.18)$$

The above relations fully characterise the scattering at a single cylindrical boundary and thus describe the response of an isolated solid polymer fibre. In a composite structure comprising many cylindrical boundaries, such as a porous fibre, each pore is represented in its own local coordinate frame. Consequently, their individual scattering matrices are not directly compatible. Coupling between cylinders is achieved by introducing a translation matrix that transforms the field expansion from one coordinate origin to another.

3.1.3 Multiple Holes

The choice of origin for expressing an electromagnetic field V is not unique. Consequently, Bessel functions may be translated between any two coordinate frames while representing the same field through Graf's Addition Theorem [43], which states that for any function $C_m \in \{J_m, Y_m, H_m^{(1)}, H_m^{(2)}\}$



$$C_m(w)e^{im\chi} = \sum_{n \in \mathbb{Z}} C_{m+n}(u)J_n(v)e^{in\alpha} . \quad (3.1.19)$$

Where u, v, w , are the side lengths and α, χ are the angles of the triangle depicted. This equation can be re-expressed in vector form, where $\arg(\vec{u})$ is the azimuthal angle of $\vec{u}$:

$$C_m(k_{\perp}|\vec{u} - \vec{v}|)e^{im \arg(\vec{u} - \vec{v})} = \sum_{n \in \mathbb{Z}} C_{m-n}(k_{\perp}|\vec{u}|)e^{i(m-n) \arg(\vec{u})}J_n(k_{\perp}|\vec{v}|)e^{in \arg(-\vec{v})} . \quad (3.1.20)$$

We require three key applications of this identity. Beginning with some known regular field centred at the origin, such as light incident from the Sun: (1) The field must be translated into the local coordinates of each cylinder so that the scattering matrices can be applied at their boundaries; (2) The scattered fields are subsequently translated from cylinder to cylinder to capture multiple scattering interactions; (3) The contributions from all cylinders are translated back to the origin and summed to yield the total scattered field. For each transformation we require change of basis matrix which converts between coefficients of the radial functions A_n^V, B_n^V .

From Origin to Cylinders

The same incoming field (e.g. from the sun) may be expressed either about a global origin or about the centre of a cylinder c . Each representation has its own coefficients, $A_n^{o(V)}, A_n^{c(V)}$, yet both describe the same physical field. The electromagnetic field at the origin is expressed

$$V = \sum_{m \in \mathbb{Z}} A_m^{o(V)}J_m(k_{\perp}|\vec{r}|)e^{im \arg(\vec{r})} . \quad (3.1.21)$$

From the Graf's Theorem vector identity Eq. (3.1.30) we make the following substitutions $C = J$, $\vec{u} - \vec{v} = \vec{r}$ and $\vec{u} = \vec{c}$ where $\vec{c}$ is the position of an arbitrary cylinder:

$$J_m(k_\perp|\vec{r}|)e^{im \arg(\vec{r})} = \sum_{n \in \mathbb{Z}} J_{m-n}(k_\perp|\vec{c}|)e^{i(m-n) \arg(\vec{c})} J_n(k_\perp|\vec{r} - \vec{c}|)e^{in \arg(\vec{r} - \vec{c})} . \quad (3.1.22)$$

Which can be substituted into the origin field basis Eq. 3.1.21:

$$V = \sum_{n,m \in \mathbb{Z}} A_m^{o(V)} J_{m-n}(k_\perp|\vec{c}|)e^{i(m-n) \arg(\vec{c})} J_n(k_\perp|\vec{r} - \vec{c}|)e^{in \arg(\vec{r} - \vec{c})} = \sum_{n \in \mathbb{Z}} A_n^{c(V)} J_n(k_\perp|\vec{r} - \vec{c}|)e^{in \arg(\vec{r} - \vec{c})} , \quad (3.1.23)$$

revealing the following relationship between the source $A_m^{o(V)}$ and cylinder $A_m^{c(V)}$ coefficients:

$$A_n^{c(V)} = \sum_{m \in \mathbb{Z}} A_m^{o(V)} J_{m-n}(k_\perp|\vec{c}|)e^{i(m-n) \arg(\vec{c})} = \sum_{m \in \mathbb{Z}} \mathcal{L}_{n,m}^{uo} A_m^{o(V)} . \quad (3.1.24)$$

Here $\mathcal{L}_{nm}^{uo}$ are the coefficients which transforms the radial functions between basis. We take a matrix representation $\mathcal{L}^{uo} \equiv [\mathcal{L}_{nm}^{uo}]$. A change of basis is required for each cylinder $c \in \{c_1, c_2 \dots c_n\}$, we therefore define the composite matrix $\tilde{L}^{co} = [\mathcal{L}^{c_1o}, \mathcal{L}^{c_2o} \dots \mathcal{L}^{c_no}]$. We also require the transformation of both polarisations of light $L^{co} = \text{diag}(\tilde{L}^{co}, \tilde{L}^{co})$. The complete transformation then can be formulated in a similar compact fashion to the scattering matrices of the previous section:

$$\mathbf{A}^c = L^{co} \mathbf{A}^o . \quad (3.1.25)$$

Here $\mathbf{A}^c = [\vec{A}^{(E)c_1} \dots \vec{A}^{(E)c_n}, \vec{A}^{(K)c_1} \dots \vec{A}^{(K)c_n}]$ and $\mathbf{A}^o = [\vec{A}^{(E)o}, \vec{A}^{(K)o}]$, where $\vec{A}^{(V)j} \equiv [A_n^{(V)j}]$. Similar definitions for $\mathbf{B}^c, \mathbf{B}^o$ will appear in the following sections. The matrix L^{co} copies and translates the source field into the basis of each cylinder.

From Cylinder to Cylinder

Once each cylinder has received the source field, scattering occurs at the cylinders boundary producing an outgoing field that impinges upon neighbouring cylinders. The outgoing field from one cylinder appears as an incoming field to another, motivating the introduction of a matrix that transforms the radial function coefficients between respective coordinate frames. Again beginning with Eq. (3.1.30), we take $C = H^{(1)}$ and set $\vec{v} = \vec{d} - \vec{r}$, with $\vec{u} = \vec{d} - \vec{c}$,

$$H_m^{(1)}(k_\perp|\vec{r} - \vec{c}|)e^{im \arg(\vec{r} - \vec{c})} = \sum_{n \in \mathbb{Z}} H_{m-n}^{(1)}(k_\perp|\vec{d} - \vec{c}|)e^{i(m-n) \arg(\vec{d} - \vec{c})} J_n(k_\perp|\vec{r} - \vec{d}|)e^{in \arg(\vec{r} - \vec{d})} . \quad (3.1.26)$$

In a similar manner to the previous change of basis, the outgoing field co-coefficients $B_m^{c(V)}$ at cylinder c may be related to the incoming field coefficients $A_n^{d(V)}$ at cylinder d :

$$A_n^{d(V)} = \sum_{m \in \mathbb{Z}} B_m^{c(V)} H_{m-n}^{(1)}(k_\perp|\vec{d} - \vec{c}|)e^{i(m-n) \arg(\vec{d} - \vec{c})} = \mathcal{L}_{n,m}^{dc} B_m^{c(V)} , \quad (3.1.27)$$

denoting the matrix form $\mathcal{L}^{dc} \equiv [\mathcal{L}_{nm}^{dc}]$ of the translation coefficients. It should be stressed that this particular change of basis is only valid in a limited region of space. Since $H_m^{(1)}(k_\perp|\vec{r} - \vec{c}|)$ represents a field radiating outward from a singularity at $\vec{c}$, the non-singular $J_n^{(1)}(k_\perp|\vec{r} - \vec{d}|)$ functions can only represent the field in a region that does not enclose the source. In practice, this transformation is applied only at cylinder boundaries, where $|\vec{r} - \vec{d}| < |\vec{d} - \vec{c}|$ and therefore the expansion remains valid. A change of basis is required between every cylinder pair $c, d \in \{c_1, c_2 \dots c_n\}$. To facilitate this we form a block matrix for all unique cylinder pairs $c \neq d$:

$$\tilde{L}^{cc} = \begin{bmatrix} 0 & \mathcal{L}^{c_1 c_2} & \dots & \mathcal{L}^{c_1 c_n} \\ \mathcal{L}^{c_2 c_1} & \ddots & & \vdots \\ \vdots & & 0 & \mathcal{L}^{c_{n-1} c_n} \\ \mathcal{L}^{c_n c_1} & \dots & \mathcal{L}^{c_n c_{n-1}} & 0 \end{bmatrix}, \quad (3.1.28)$$

Again forming $L^{cc} = \text{diag}(\tilde{L}^{cc}, \tilde{L}^{cc})$ to transform both polarisations and produce the complete relation

$$\mathbf{A}^c = L^{cc} \mathbf{B}^c. \quad (3.1.29)$$

This equation formalises the mutual coupling between cylinders, expressing that the field incident on one cylinder arises from the sum of many outgoing fields emerging from neighbouring cylinders.

From Cylinders to Origin

Finally, we return to our original basis. Beginning with Graf's Theorem Eq. (3.1.30), take $C = H^{(1)}$ and set $\vec{u} = \vec{r}$ and $\vec{v} = \vec{c}$:

$$H_m^{(1)}(k_\perp|\vec{r} - \vec{c}|)e^{im \arg(\vec{r} - \vec{c})} = \sum_{n \in \mathbb{Z}} H_{m-n}^{(1)}(k_\perp|\vec{r}|)e^{i(m-n) \arg(\vec{r})} J_n(k_\perp|\vec{c}|)e^{in \arg(-\vec{c})}. \quad (3.1.30)$$

This permits the transformation of the outgoing field coefficients $B_m^{c(V)}$ at cylinder c to outgoing field coefficients $B_n^{o(V)}$ in the origin basis :

$$B_n^o = \sum_{m \in \mathbb{Z}} (-1)^{n-m} J_{n-m}(k_\perp|\vec{c}|)e^{i(n-m) \arg(\vec{c})} B_m^c = \mathcal{L}_{nm}^{oc} B_m^c, \quad (3.1.31)$$

defining the matrix form $\mathcal{L}^{oc} = [\mathcal{L}_{nm}^{oc}]$. This transformation is only valid outside the cylindrical boundary enclosing the original source $|\vec{r}| > |\vec{c}|$. The restriction arises because both sets of Hankel functions approach infinity at their respective centres $|\vec{r}| = 0$ and $\vec{r} = |\vec{c}|$. The new expansion will only converge in a region where neither singularity lies. This expansion is employed to evaluate scattering matrices at the fibre boundary, where all holes lie within $|\vec{c}_i| < a_f$. A transformation to the origin is required for every cylinder $c \in \{c_1, c_2 \dots c_n\}$ motivating the definition of the block matrix

$$\tilde{L}^{oc} = [\mathcal{L}^{oc_1 T}, \mathcal{L}^{oc_2 T} \dots \mathcal{L}^{oc_n T}]^T, \quad (3.1.32)$$

where T represents the transpose. Accounting for both polarisations $L^{oc} = \text{diag}(\tilde{L}^{oc}, \tilde{L}^{oc})$ yields

$$\mathbf{B}^o = L^{oc} \mathbf{B}^c \quad (3.1.33)$$

Which encapsulates the translation and sum of many outgoing cylindrical fields collapsed into a single outgoing field emanating from the origin.

3.1.4 The Complete Scattering Matrix

Armed with this arsenal of matrices we now return to the full geometry of Fig. 3.1. The complete scattering behaviour of the porous fibre can be expressed in terms of a global scattering matrix, which takes into account the multiple scattering between holes as well as scattering from and through the external boundary of the fibre. The scattering matrix defines the transformation between incoming and outgoing electromagnetic field coefficients at the boundary of a general porous fibre:

$$\mathbf{B}_e^o = \mathbf{S} \mathbf{A}_e^o. \quad (3.1.34)$$

The subscript denotes the region a particular field is in, fields inside the fibre will be denoted with an f subscript and those external to the fibre an e subscript. The superscript will denote the basis of the field, for example: o the origin and c the cylinders. At the fibre boundary the external outgoing field is a composition of the reflected source field $\mathbf{A}_e^o$ combined with some transmission of the internal outgoing fields $\mathbf{B}_f^c$ originating from the scattering holes:

$$\mathbf{B}_e^o = T_e^o L^{oc} \mathbf{B}_f^c + R_e^o \mathbf{A}_e^o. \quad (3.1.35)$$

We must now derive the relationship for the unknown internal outgoing field $\mathbf{B}_f^c$ with respect to the incoming field $\mathbf{A}_e^o$. The outgoing field for a single cylindrical hole is related to the incoming field by the following expression:

$$\mathbf{B}_f^{c1} = R_f^{c1} \mathbf{A}_f^{c1}. \quad (3.1.36)$$

Each cylinder has its own unique set of scattering matrices $\{R_f^{c1}, R_{c1}^{c1}, T_f^{c1}, T_{c1}^{c1}\}$ which depend on the size and index contrast of the cylinder boundary. We denote the combined reflection matrix for every cylinder $R_f^{c(E E)} = \text{diag}(R_f^{c1(E E)}, R_f^{c2(E E)} \dots R_f^{cn(E E)})$, such that a complete matrix for both polarisations can be constructed

$$\mathbf{R}_f^c = \begin{bmatrix} R_f^{c(E E)} & R_f^{c(E K)} \\ R_f^{c(K E)} & R_f^{c(K K)} \end{bmatrix}. \quad (3.1.37)$$

Through this matrix, the field transformation and cross coupling between incoming and outgoing coefficients is represented:

$$\mathbf{B}_f^c = \mathbf{R}_f^c \mathbf{A}_f^c \quad (3.1.38)$$

With similar constructions for $\mathbf{R}_e^c, \mathbf{T}_f^c, \mathbf{T}_e^c$. In Eq. 3.1.38, the incoming field $\mathbf{A}_f^c$ has many origins, which may be expressed distinctly:

$$\mathbf{B}_f^c = \mathbf{R}_f^c (L^{cc} \mathbf{B}_f^c + L^{co} R_f^o L^{oc} \mathbf{B}_f^c + L^{co} T_f^o \mathbf{A}_e^o). \quad (3.1.39)$$

In this equation, the first term in the brackets $L^{\text{cc}}\mathbf{B}_f^{\text{c}}$ represents the fields coming from the scattering of other holes. The second term represents the internally reflected component (R_f^{o}), produced when the field scattered by all holes collectively reflects off the outer fibre boundary. The final term represents sources outside the fibre transmitted (T_f^{o}) through the boundary. Internal reflections require two change of basis as the field propagates from the cylinders to the edge of the fibre and returns to the cylinders. Notably, this relation contains $\mathbf{B}^{\text{c}}$ on both sides, indicating that the scattered cylinder term must therefore be solved self consistently. Leading to

$$\mathbf{B}_f^{\text{c}} = [I - \mathbf{R}_f^{\text{c}}(L^{\text{cc}} + L^{\text{co}}R_f^{\text{o}}L^{\text{oc}})]^{-1}\mathbf{R}_f^{\text{c}}L^{\text{co}}T_f^{\text{o}}\mathbf{A}_e^{\text{o}}, \quad (3.1.40)$$

where I denotes the identity matrix. Since all R, T and L matrices are known analytically, Eq. 3.1.40 gives a direct expression for the outgoing fields inside the fibre for an arbitrary incoming field $\mathbf{A}_e^{\text{o}}$. However, it requires the inversion of a potentially large matrix, which is the primary computational bottleneck. Several alternatives exist to avoid direct inversion, though each introduces trade-offs in accuracy or convergence (more on this in Chapter 4). Once we have solved for the internal outgoing field $\mathbf{B}_f^{\text{c}}$, we return to the original equation 3.1.35 and substitute $\mathbf{B}_f^{\text{c}}$:

$$\mathbf{B}_e^{\text{o}} = T_e^{\text{o}}L^{\text{oc}}[I - \mathbf{R}_f^{\text{c}}(L^{\text{cc}} + L^{\text{co}}R_f^{\text{o}}L^{\text{oc}})]^{-1}\mathbf{R}_f^{\text{c}}L^{\text{co}}T_f^{\text{o}}\mathbf{A}_e^{\text{o}} + R_e^{\text{o}}\mathbf{A}_e^{\text{o}}, \quad (3.1.41)$$

yielding the global scattering matrix $\mathcal{S}$:

$$\mathcal{S} = T_e^{\text{o}}L^{\text{oc}}[I - \mathbf{R}_f^{\text{c}}(L^{\text{cc}} + L^{\text{co}}R_f^{\text{o}}L^{\text{oc}})]^{-1}\mathbf{R}_f^{\text{c}}L^{\text{co}}T_f^{\text{o}} + R_e^{\text{o}}. \quad (3.1.42)$$

This matrix contains the complete electromagnetic description of any porous fibre. Once this matrix is known, the scattered field for *any* incoming field can be calculated through simple matrix-vector multiplication.

3.2 Deriving the Scattering Power

In the previous section we derived the scattering matrix $\mathcal{S}$ that transfers between known incident Bessel coefficients to the scattered unknown outgoing Hankel coefficients

$$B_n^{\text{V}} = \sum_m \mathcal{S}_{n,m}^{\text{VV}'} A_m^{\text{V}'}. \quad (3.2.1)$$

Here we have shifted from a matrix representation to indexed summation and separated the polarisations. For passive cooling we are particularly interested in the intensity of light scattered by a porous fibre. The scattering depends greatly on the position and size of each hole, and on wavelength and angles of incidence. For daytime radiative cooling, we ultimately want to minimise any sunlight going through the fibre to the skin, instead maximising the backscattered power, across the entire solar spectrum and across all angles of incidence. The time-averaged radiated intensity is obtained from the Poynting vector [44]

$$\langle \vec{S} \rangle = \frac{1}{2Z_0} \Re(\vec{E} \times \vec{K}^*). \quad (3.2.2)$$

where $\Re$ denotes the real component and Z^0 the vacuum impedance. For an infinitely long cylinder the total intensity scattered is captured solely by the radial component:

$$\langle \vec{S}_r \rangle = \frac{1}{2Z_0} \Re(E_\theta K_z^* - E_z K_\theta^*) \hat{r} . \quad (3.2.3)$$

The azimuthal field components E_θ can be expressed in terms of the longitudinal E_z components using Eq. (3.1.4). In the far field these relationships simplify to

$$E_\theta \approx -\frac{ik_0}{k_\perp^2} \frac{\partial K_z}{\partial r} \quad K_\theta \approx \frac{ik_0}{k_\perp^2} \frac{\partial E_z}{\partial r} . \quad (3.2.4)$$

Substituting the Fourier-Bessel expansion of the fields, it may then be shown (see Appendix B) that

$$\langle S_r \rangle = \frac{-k_0}{Z_0 k_\perp^2 \pi r} \sum_{n,m \in \mathbb{Z}} \Re \left[\left(\frac{1}{4} (A_n^E A_m^{E*} + A_n^K A_m^{K*}) [(-1)^{n-m} - 1] \right. \right. \\ \left. \left. - \Re(A_n^E B_m^{E*} + A_n^K B_m^{K*}) - B_n^E B_m^{E*} - B_n^K B_m^{K*} \right) e^{i(n-m)(\theta - \frac{\pi}{2})} \right] . \quad (3.2.5)$$

3.2.1 Power in the Forward Direction

Minimizing the forward transmitted power of a single fibre is critical for passive radiative cooling. Forward scattering allows incident solar radiation to continue through the material, contributing to unwanted heating. In contrast, strong backward or large angle scattering redirects sunlight away from the fibre, enhancing solar reflectance and improving cooling efficiency. Directional control is especially important in multiple scattering systems, such as porous or fibrous ensembles. Light interacts with many cylinders sequentially producing a compounded suppression of light in the forward direction.

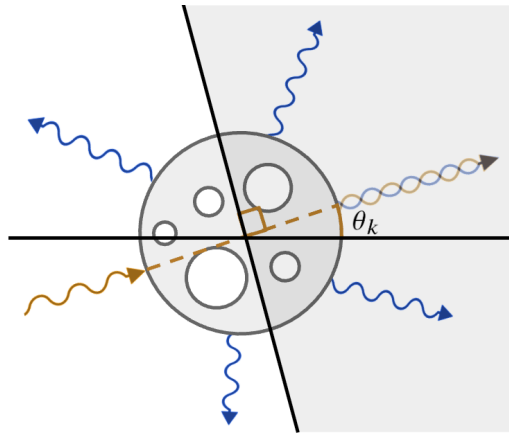


Figure 3.3: Incident light (Orange) arrives at an angle θ_k , Scattered Light (Blue) is radiated in all directions. Scattered light interferes with the incident field. The shaded region is the forward half plane, we desire minimal power transfer in this region.

To quantify the forward power we take an azimuthal integral over the half plane either side of the incident light direction θ_k (Fig. 3.3):

$$\int_{\theta_k - \pi}^{\theta_k + \pi} \langle S_r \rangle r d\theta. \quad (3.2.6)$$

Evaluation of this integral yields the power radiated in the forward direction per unit length, where $\text{sinc } x = \frac{\sin x}{x}$:

$$P_f = \frac{-k_0}{Z_0 k_{\perp}^2} \sum_{n,m \in \mathbb{Z}} \Re \left[\left(\frac{1}{2} (A_n^E A_m^{E*} + A_n^K A_m^{K*}) [(-1)^{n-m} - 1] \right. \right. \\ \left. \left. - \Re(A_n^E B_m^{E*} + A_n^K B_m^{K*}) - B_n^E B_m^{E*} - B_n^K B_m^{K*} \right) e^{i(n-m)(\theta_k - \frac{\pi}{2})} \text{sinc} \left((n-m) \frac{\pi}{2} \right) \right] \quad (3.2.7)$$

The power is entirely dependant on the source A_n^V and scattered coefficients B_n^V which are related by the scattering matrix. The source coefficients are known analytically and can now be introduced. The source field coming from the sun is approximated as a plane wave travelling in the azimuthal direction θ_k . The light impinges on the cylindrical fibre at an oblique angle ϕ_k relative to the longitudinal z-axis such that a small ϕ_k corresponds to grazing incidence. The oblique angle of light determines the particular k_z component excited within the fibre $k_z = k_0 \cos(\phi_k)$ where k_0 is the vacuum wave number. The parameter δ denotes the polarisation angle depicted in Fig. 3.4.

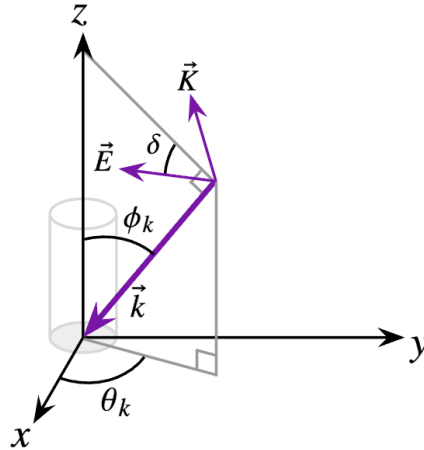


Figure 3.4: Coordinate system and angular definitions for a plane wave incident on a cylindrical fibre.

The incident longitudinal fields can therefore be expressed as

$$E_z = V_0 \sin(\phi_k) \cos(\delta) e^{ik_{\perp} r \cos(\theta - \theta_k)}, \quad K_z = V_0 \sin(\phi_k) \sin(\delta) e^{ik_{\perp} r \cos(\theta - \theta_k)} \quad (3.2.8)$$

Where V_0 is the amplitude of the incident plane wave. The fields scattered by each hole are described in a cylindrical basis, so the incident plane wave must be brought into the same form. This is achieved through the Jacobi–Anger expansion, which expresses the plane wave as a weighted sum of Bessel functions [45]:

$$e^{iz \cos \theta} = \sum_{m \in \mathbb{Z}} i^m J_m(z) e^{im\theta}. \quad (3.2.9)$$

Substitution into Eq. (3.2.8) yields

$$A_m^E = V_0 \sin(\phi) \cos(\delta) e^{im(\frac{\pi}{2} - \theta_k)} \quad A_m^K = V_0 \sin(\phi) \sin(\delta) e^{im(\frac{\pi}{2} - \theta_k)} \quad (3.2.10)$$

Illumination of a textile is inherently multi-directional: sunlight reaches each fibre from a range of incident angles, and in a bulk material, each fibre has a random orientation. Therefore to accurately quantify the general behaviour of a single fibre the forward power should be averaged over all angles of illumination. Here it was realised that the multipole formalism facilitates a dramatic computational speed up. In a typical solver one would discretize the incident angles of solar radiation θ_k, ϕ_k , simulate with the corresponding source coefficients for each set of sample angles and numerically integrate the forward power obtained from many solves. In the multipole formalism, the scattering matrix already includes the information for azimuthal scattering θ_k for *any* incoming field. Once the matrix $\mathcal{S}$ is computed, it should thus be possible to extract the average forward scattering for all angles of incidence from the matrix coefficients. The method sidesteps the brute force angular sweeps required by finite difference solvers, yielding a considerable speed and greater accuracy. This new derivation is detailed in Appendix B and the result is a closed form expression of the azimuthally averaged forward power:

$$\langle P_f \rangle_\theta = \frac{-V_0^2 \tan^2(\phi_k)}{2Z_0 k_0} \sum_{m,n,p,q \in \mathbb{Z}} \Re \left[\left(\frac{1}{4} ((-1)^{n-m} - 1) - \Re(\mathcal{S}_{m,m}^{EE} + \mathcal{S}_{m,m}^{KK}) \right. \right. \\ \left. \left. - (\mathcal{S}_{n,q}^{EE} \mathcal{S}_{m,p}^{*EE} + \mathcal{S}_{n,q}^{EK} \mathcal{S}_{m,p}^{*EK} + \mathcal{S}_{n,q}^{KE} \mathcal{S}_{m,p}^{*KE} + \mathcal{S}_{n,q}^{KK} \mathcal{S}_{m,p}^{*KK}) \delta_{n-m,q-p} \right) \text{sinc}((n-m)\frac{\pi}{2}) \right]. \quad (3.2.11)$$

However, integrating the intensity of an ideal plane wave over an infinite half plane yields infinite power, and thus a divergent forward power. To allow for meaningful interpretation, we need to compare this forward power with the power that would have been going forward in the absence of the fibre. To obtain a finite result, the divergent first term in Eq. 3.2.11 is subtracted. This term represents the power of the incident source, as it is the only contribution that remains when no scattering occurs ($B_n^V = 0$). This yields the power difference $\langle \Delta P_f \rangle_\theta$, and we normalise this quantity by the power of the incident plane wave and fibre diameter to arrive at the figure of merit:

$$\langle F \rangle_\theta = \frac{1}{2k_\perp a} \sum_{m \in \mathbb{Z}} \Re [2(\mathcal{S}_{m,m}^{EE} + \mathcal{S}_{m,m}^{KK}) + \\ \sum_{p,d} (\mathcal{S}_{m+d,p+d}^{EE} \mathcal{S}_{m,p}^{*EE} + \mathcal{S}_{m+d,p+d}^{EE} \mathcal{S}_{m,p}^{*EE} + \mathcal{S}_{m+d,p+d}^{EE} \mathcal{S}_{m,p}^{*EE} + \mathcal{S}_{m+d,p+d}^{EE} \mathcal{S}_{m,p}^{*EE}) \text{sinc}(\frac{d\pi}{2})] . \quad (3.2.12)$$

The quantity $\langle F \rangle_\theta$ is a dimensionless measure of the incident power redirected or extinguished in the half plane defined by the propagation direction of an incoming plane wave (Fig 3.3). The first term represents interference between the incident and scattered fields, while the second term accounts for forward scattered light. Together, these terms capture the total modification to the forward intensity resulting from both interference and direct scattering. The second term in the expression is a triple sum, but can be computed with remarkable efficiency using a technique outlined in Chapter 4. To provide intuition for the metric, we plot in Figure 3.5 the real component of the Electric field alongside

the figure of merit: for a single cylinder containing a vacuum (i.e. no cylinder), dielectric and a perfect conductor.

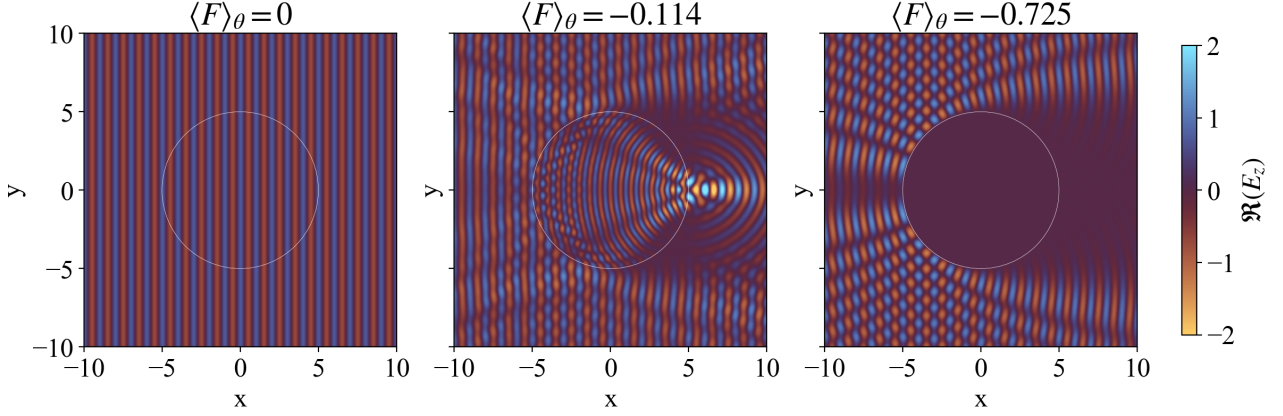


Figure 3.5: An $\vec{E}$ polarisation ($\delta = 0$) plane wave with wavelength $\lambda = 1\mu\text{m}$ propagates in the $+x$ direction $\theta_k = 0$ at normal incidence $\phi_k = 0$. The plane wave scatters off a cylindrical fibre of radius $a = 5\mu\text{m}$ with varying refractive index: (left) $n = 1 + 0i$ (center) $n = 1.5 + 0i$ (right) $n = 1 + \infty i$

A fibre with refractive index $n = 1$ does not induce scattering and therefore the forward difference $\langle F \rangle_\theta$ is strictly zero. The introduction of a solid polymer fibre $n = 1.5$ marginally increases scattering and destructive interference with a forward difference $\langle F \rangle_\theta = -0.114$. A perfect conductor casts a large shadow in the direction of light propagation yielding a strong forward difference $\langle F \rangle_\theta = -0.725$. To determine whether the chosen fibre radius has converged in the limit of geometric optics, we plot $\langle F \rangle_\theta$ as a function of conductor radius a_f .

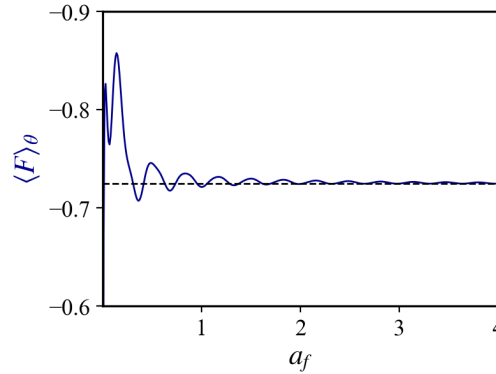


Figure 3.6: Normalised forward power difference of a cylindrical perfect conductor as a function of conductor radius a_f , $\lambda = 1\mu\text{m}$, $n = 1 + \infty i$, at normal incidence $\phi_k = 0$

The convergence of $\langle F \rangle_\theta \rightarrow -0.725$ confirms that the fibre radius $a_f = 5\mu\text{m}$ is large enough for the geometric optics regime, and can therefore serve as a reference for scattering strength. This value will be used to benchmark porous fibres in Chapter 5. There, we will explore how the addition of holes into a polymer fibre can boost the forward difference to a magnitude closer to that of a perfect conductor. A solar absorptive polymer fibre could also strongly reduce the forward propagating power, but would do so by converting optical energy into heat, which is undesirable for radiative cooling applications. Absorption and emission become particularly relevant in the infrared, where polymer losses enable re-radiation of heat into the cold universe. Fibre absorption and emissivity could be

incorporated into a more general figure of merit; however, in this work we restrict the analysis to the solar spectrum, where the majority of textile polymers behave approximately as lossless dielectrics.

This chapter presented the generalised multipole method for modelling fibres containing many cylindrical holes and demonstrated how their collective scattering can be expressed through a global scattering matrix. From this matrix, a new figure of merit: the angle averaged forward difference was derived to quantify the strength of solar scattering. In the next chapter, the numerical implementation of the multipole method and figure of merit is presented in detail, along with an assessment of accuracy and computational performance.

4 Implementation

Computational efficiency is central to this work: identifying an optimal fibre requires solving the full multipole scattering problem for thousands of geometries across multiple wavelengths and incident angles. The numerical implementation developed in this work consists of two main components. A multipole scattering module that performs the field calculations, and an optimisation loop that refines the fibre geometry using gradient based updates. The solver and optimisation routines together comprise nearly a thousand lines of code, with accompanying Jupyter notebooks used for analysis extending far beyond this. The primary code developed for this project has been compiled into a public GitHub repository (available at github.com/danielglass1/Daniel-Glass-Honours). The folder, `mp_torch`, contains the components of the multipole solver and the file `optimiser` is the optimisation routine. An example script using the solver is provided in Appendix C.

4.1 The Solver

The multipole method described in Chapter 3 forms the basis of the scattering engine. Its analytic representation of fields and boundary conditions through Bessel and Hankel functions makes the solver inherently differentiable, allowing gradients of the scattering function $\langle F \rangle_\theta$ to be obtained with respect to any input parameters. Care must be taken in how these derivatives are evaluated. Symbolic differentiation is not feasible: the multipole method involves thousands of Bessel functions, translation operators, and large matrix inversions. Differentiating these expressions symbolically leads to mammoth sized algebraic expansions that grow combinatorially before simplification resulting in memory overflow. Finite differences are equally inefficient: to estimate a gradient, every parameter must be individually perturbed and the entire scattering problem resolved. For a fibre with twenty holes, sixty solves must occur per iteration for the x , y position and radius of each hole. This introduces both extreme computational cost and step size error. A cunning approach is required to sidestep both algebraic overhead and repeated evaluations.

4.1.1 Automatic Differentiation

Automatic differentiation computes exact derivatives at a single point in the input space, rather than producing a general symbolic expression. This is sufficient for optimisation, which relies solely on the gradient at the current geometry to guide parameters towards optima. This is achieved by decomposing

a function into a sequence of elementary operations: addition, multiplication, trigonometric functions, each with an analytic derivative. As a program executes, primitive operations are recorded in a graph that mirrors the structure of the code. Each node in this graph stores the partial derivative of the output with respect to the inputs. When a derivative is requested, the program traverses backward through the computation graph, numerically applying the chain rule to return the derivative at the input point (see Fig. 4.1).

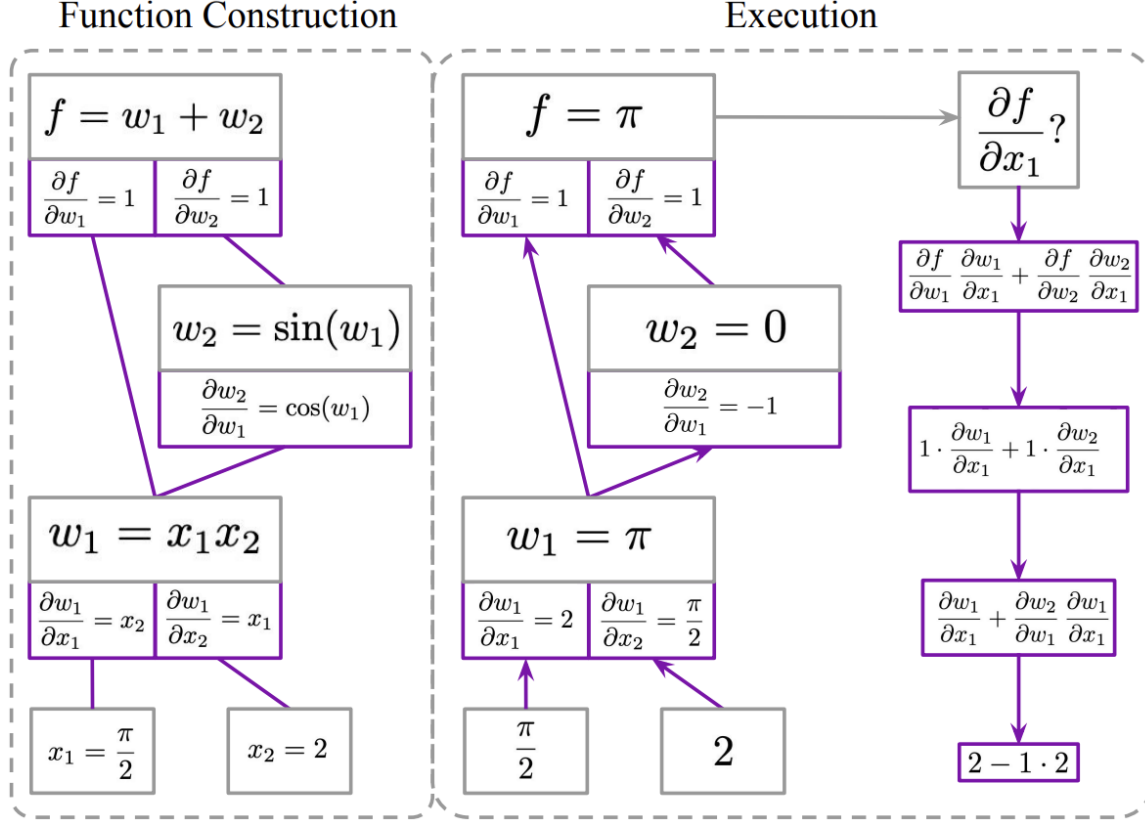


Figure 4.1: Illustration of automatic differentiation. The programmer implements the function as usual, while the automatic differentiation library constructs a computation graph. When a derivative is called the chain rule is applied in reverse to obtain the local derivative at the input point $(\frac{\pi}{2}, 2)$.

Several automatic differentiation libraries were considered, including *JAX* [46], *PyTorch* [47], and *TensorFlow* [48]. Ultimately *PyTorch* was chosen for its flexible dynamic graph, straightforward debugging, and library maturity. All available libraries lacked native support for cylindrical Bessel and Hankel functions of arbitrary order. To enable automatic differentiation through the multipole solver, we developed a set of wrapped *SciPy* functions `scipy.special.jv`, `yv`. Which were implemented within `torch.autograd.Function` classes, with derivatives defined by the Bessel recurrence relations $2C'_n = C_{n-1} - C_{n+1}$ $C \in \{J_n, Y_n, H_n^{(1)}, H_n^{(2)}\}$ [43] to enable the backward pass. While this approach allowed seamless integration with automatic differentiation, the overhead of repeatedly calling *SciPy* remains. Future work could implement native Bessel and Hankel functions in *PyTorch* alongside a lookup dictionary of pre-computed function values to be reused across optimisation. Gradients obtained from automatic differentiation were compared against finite difference estimates for numerical validation and agreed to near machine precision.

Having established a framework for efficient gradient computation, attention was directed towards

improving the efficiency of the solver itself.

4.1.2 Performance Engineering the Solver

The *PyTorch* library is built on a C++ back-end which supports efficient tensor operations and optional GPU acceleration. The solver operates by default in 32 bit precision, required by *PyTorch*'s CUDA backend for GPU compatibility. While all calculations in this paper were run on CPU, the implementation remains GPU supported for future scalability and can be executed in 64 bit precision when higher accuracy is required. Explicit Python loops were replaced with vectorised *PyTorch* tensor operations, reducing interpreter overhead and reducing runtime.

The principal computational bottleneck arises from the repeated inversion of large dense matrices 3.1.40 that couple multipole orders between cylinders. Inversion using LU decomposition scales cubically with matrix dimension, dominating optimisation runtime where inversion is required for each geometry update. Several iterative methods were explored as alternatives to LU decomposition, mirroring the physical process of successive scattering events where fields are repeatedly exchanged between cylinders until convergence. A generalized minimal residual (GMRES) algorithm [49] was implemented with the package *xitorch* [50], but found to take longer to converge than LU matrix inversion. The dense multipole matrices, combined with the absence of an effective pre-conditioning matrix and the lack of a native *PyTorch* C++ back-end contributed to its inadequacy. A Neumann series expansion [51] was also tested as an approximate inverse, but often failed to converge as the matrix spectral radius exceeded unity for many geometries. For optimisation, however, where each gradient update perturbs the geometry only slightly, it may be possible to iteratively estimate the inverse of the matrix difference, $\Delta M = M_{i+1} - M_i$, which is small relative to the matrix itself. Such an approach could yield stable convergence through a Neumann iteration, though accumulated numerical error and the need for damping remain challenges that are not addressed in this work.

Another lengthy computation is the summation over angular orders m, p, d required for the figure of merit $\langle F \rangle_\theta$ (see Eq. 3.2.12) evaluates scattering matrix values weighted by a sinc function:

$$\sum_{m,p,d \in \mathbb{Z}} \text{sinc}\left(\frac{d\pi}{2}\right) \mathcal{S}_{m+d,p+d}^{EE} \mathcal{S}_{m,p}^{EE*}. \quad (4.1.1)$$

However, we note that the sum contains the 2D discrete auto-correlation $C[d]$:

$$\sum_{d \in \mathbb{Z}} C[d] \text{sinc}\left(\frac{d\pi}{2}\right). \quad (4.1.2)$$

Remarkably, the Wiener–Khinchin theorem [52] states that the autocorrelation $C[d]$ is equivalent to the following

$$C[d] = \mathcal{F}\{|\mathcal{F}^{-1} \mathcal{S}^{EE}|^2\}, \quad (4.1.3)$$

where $\mathcal{F}$ denotes the Fourier transform. Implementation with the Fast Fourier Transform reduces the computational cost of the triple sum from $\mathcal{O}(N^3)$ to $\mathcal{O}(N^2 \log N)$. Convolution based acceleration was also tested for the Graf translation matrices, which share a similar structure, but these proved numerically unstable and warrant further investigation.

The numerical implementation developed in this chapter was validated across a range of asymmetric cylindrical geometries under both normal and oblique incidence with complex refractive indices. To validate the solver under normal incidence we compared scattering cross-sections and outgoing Hankel coefficients with MieSolver [53], an established object-oriented Mie-series code for cylindrical scattering. For oblique incidence, scattering cross-sections and coefficients were compared with CUDOS MOF Utilities [1], showing agreement to within numerical precision.

Having forged a rapid solver, we are now equipped to tackle optimisation.

4.2 Optimisation

The objective of the optimisation routine is to locate porous fibre geometries that minimise the forward difference metric defined in Chapter 3. An arbitrary geometry contains N holes and is described by a set of parameters:

$$\mathbf{q} = \{x_i, y_i, a_i\}_{i=1}^N \quad (4.2.1)$$

where (x_i, y_i) denote the position and a_i denotes the radius of the i^{th} hole. The optimisation problem involves numerous non-linear geometric constraints: each hole must remain within the fibre boundary and must not overlap with others. Formalising these constraints, we define b_i as

$$b_i(\mathbf{q}) = \sqrt{x_i^2 + y_i^2} + a_i - a_{fibre}, \quad (4.2.2)$$

and c_{ij} as

$$c_{ij}(\mathbf{q}) = a_i + a_j - \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (4.2.3)$$

So that the optimization problem can be expressed as follows:

$$\begin{aligned} \min_{\mathbf{q}} \quad & \langle F(\mathbf{q}) \rangle_{\theta} \\ \text{subject to:} \quad & b_i(\mathbf{q}) < 0 \\ & c_{ij}(\mathbf{q}) < 0 \quad \forall i < j. \end{aligned} \quad (4.2.4)$$

For N holes, we have in total $N(N + 1)/2$ non-linear inequalities. Non-linear inequality constraints are computationally expensive to enforce and *PyTorch* gradient descent modules operate only on unconstrained objectives. The constraints are therefore relaxed into a penalty function and the optimisation reformulated as

$$\min_{\mathbf{q}} \quad \langle F(\mathbf{q}) \rangle_{\theta} + \alpha P(\mathbf{q}), \quad (4.2.5)$$

where the penalty function is defined as

$$P(\mathbf{q}) = \sum_i \text{softplus}(b_i(\mathbf{q})) + \sum_{i < j} \text{softplus}(c_{ij}(\mathbf{q})). \quad (4.2.6)$$

Here, α determines the strength of the overlap penalty, the function $\text{softplus}(x) = \frac{1}{\beta} \ln(1 + e^{\beta x})$ is a smooth, differentiable approximation to the standard ReLU function $\text{ReLU}(x) = \max(0, x)$, outputting

zero when its argument is negative and increases linearly for positive inputs. This ensures that no penalty is applied when holes do not overlap and that the penalty grows proportionally to the amount of overlap. The parameter β controls the sharpness of the transition from 0 to x for the softplus function and can be adjusted alongside α to tune the leniency of the overlap penalty. It was found for a small number of holes $0 - 10$ $\alpha \approx 1$ was appropriate where for larger hole counts a smaller $\alpha \approx 0.5$ was preferred, indicating some need for penalty normalisation. The β parameter controls how close boundaries can approach before contributing to the penalty and can be adjusted according to manufacturing requirements, such as maintaining a minimum spacing between fibres to avoid structural failure. Here, we found $\beta = 25$ permitted the cylinders to approach closely while keeping the optimisation stable, higher values making derivatives unreliable.

With the optimisation problem stated, we next generate pore geometries to seed the gradient descent process. Random fibre cross-sections were generated using a custom circle-packing algorithm that populates the fibre with non-overlapping circular holes. Random circle radii are first sampled from a beta distribution [54]. This distribution is chosen because it produces bounded random values with a controllable central tendency and adjustable spread, thereby tuning the overall filling fraction. These randomly generated radii sorted in descending order so that larger holes are placed first. Each circle is positioned by drawing random coordinates and testing for overlap until a valid location is found or a cutoff is reached.

To avoid poor local minima, many seed geometries were generated and screened to retain the best candidate before gradient descent. This was particularly effective for low hole counts, where the parameter space is less high dimensional. Gradient based optimisation then proceeds until the improvement in the figure of merit falls below 10^{-3} for several iterations.

The gradient descent algorithm employed was AdamW [55], which was found to outperform SGD and Adam. AdamW allows each parameter to adjust its step size based on its own gradient history alongside momentum to ensure progress despite over a highly oscillatory objective function.

Evolutionary optimisation algorithms were also explored for geometry pre-screening but proved inefficient. Their parameter swapping and mutation steps often produced overlapping invalid geometries with an encoding based on hole radii and positions, resulting in a large number of wasted solver evaluations. Different encodings such as hole separation or higher order statistical features could benefit from further exploration. We did not encode by pairwise hole separation because this inflates the parameter space from $O(N)$ to $O(N^2)$ and introduces more geometric constraints such as triangle inequalities.

With a robust solver and optimisation framework in place, we can now extract physical insight.

5 Numerical Results

This chapter examines how porous fibre structure controls the angular distribution of scattered light and, by extension, its cooling capability. The azimuthally averaged forward difference $\langle F \rangle_\theta$ is the key measure used throughout this work to quantify how a fibre affects solar light propagation. It represents the reduction in forward scattered power due to fibre scattering, averaged over any azimuthal incident angle. More negative values are indicative of stronger backward scattering and destructive interference in the forward direction, resulting in less solar energy transmitted through the material.

When combined with mid-infrared emission, this behaviour directly enhances the fibre's cooling performance. This chapter aims to analyse the forward scattering in fibres of varying levels of complexity. We will begin with the simplest, single hole geometry at a single wavelength as a benchmark, and eventually extend to broadband optimised multi hole fibres.

5.1 Monochromatic Scattering

The solar spectrum ranges predominantly between $\lambda = 0.3\text{-}2.5\ \mu\text{m}$ [56], peaking in the visible light region. Typical fibre diameters fall between $10\text{ - }20\ \mu\text{m}$, a natural fibre such as silk has a diameter of approximately $12\ \mu\text{m}$ [57]. In the following sections the fibre radii denoted a_f will be set to $5\ \mu\text{m}$ and the incident wavelength to $1\ \mu\text{m}$ to ensure a representative optical scale while maintaining computational efficiency. Larger fibres necessitate higher Bessel truncation orders to capture the field accurately, substantially increasing the simulation cost.

5.1.1 Single Hole

The figure of merit $\langle F \rangle_\theta$ quantifies the fibre's impact on forward power propagation averaged over azimuthal incidence. Consequently any azimuthal rotation of a fibre's geometry produces the same value of $\langle F \rangle_\theta$. A fibre of fixed radius with a single hole may therefore be parametrised by two values, hole position with respect to the fibre centre x_h and hole radii a_h pictured in Fig. 5.1.

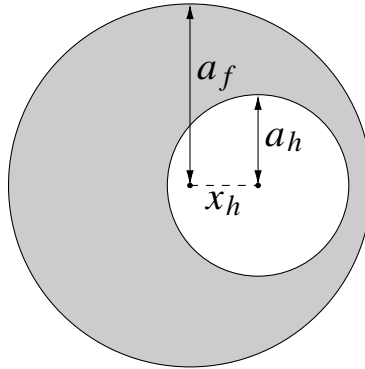


Figure 5.1: Geometry of a single-hole fibre, defined by the fibre radius a_f , hole radius a_h , and hole displacement x_h from the fibre centre.

The forward difference metric $\langle F \rangle_\theta$ was computed over a range of hole position x_h and radius a_h in Fig 5.2. The triangular region marks valid, non-overlapping configurations where $x_h + a_h \leq a_f$. The function is highly oscillatory. Oscillations predominantly arise from Mie resonances where the hole diameter approaches Bessel function roots in the scattering matrix. Two separate regions of strongly reduced forward scattering emerge: (1) a large centred hole leaving a thin shell (2) an off-centre hole near the very edge of the fibre. When the hole is displaced from the centre $x_h \neq 0$, scattering becomes azimuthally dependent, enhanced along some directions and suppressed in others. The centred configuration (1) exhibited the strongest resonant extrema, with deeper minima and greater maxima than off-centred case (2). This behaviour motivated a separate plot focused on a centred hole $x_h = 0$, (Figure 5.2 right). From this plot, the optimal fibre appears to be for

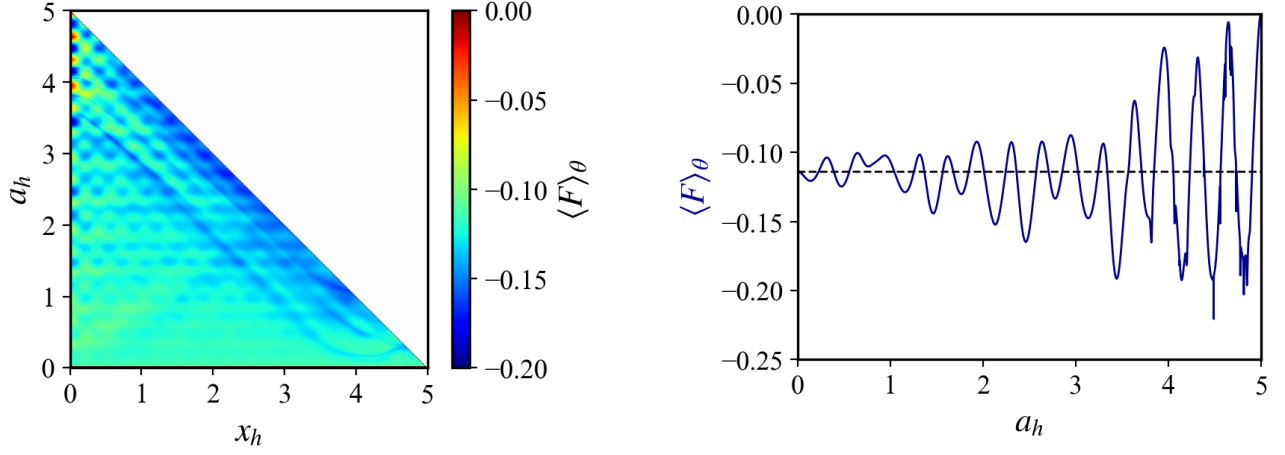


Figure 5.2: *Left:* Forward difference $\langle F \rangle_\theta$ as a function of hole position x_h and radius a_h . *Right:* Slice of left plot, $\langle F \rangle_\theta$ as a function of hole radius a_h for a centred hole $x_h = 0$.

$a_h \simeq 4.5$. Gradient based optimisation found a minimum of $\langle F \rangle_\theta = -0.220$ at $x_h \approx 0, r_h \approx 4.49\mu\text{m}$ matching the minima observed in the Fig. 5.2. The optimal configuration corresponds to a half wavelength thickness symmetric hollow fibre, in which the polymer shell resembles a Fabry P  rot cavity supporting constructive interference and strong reflection near the outer boundary. The introduction of a single central hole leads to a dramatic improvement over the solid fibre baseline of $\langle F \rangle_\theta = -0.114$, demonstrating the strong influence of internal air inclusions on scattering performance. Nevertheless, the result remains well below the perfect conductor benchmark of $\langle F \rangle_\theta = -0.725$ discussed in Chapter 3. To further enhance the forward suppression, additional holes can be introduced to exploit multiple scattering.

5.1.2 Many Holes

Each hole introduces new degrees of freedom in both radius and position. Collectively, these parameters form a high dimensional space that is difficult to map out visually and computationally. Consequently, random sampling and gradient optimisation become the primary method of exploring the design landscape. First, to get an overview of what aspect of the geometry might be important, we calculate $\langle F \rangle_\theta$ for a random set of fibre realizations and consider the correlation with simple statistic measures such as the air filling fraction and hole count. Then we proceed to gradient optimisation to identify optimal geometries that outperform their randomly generated counterparts.

Random Sampling

In the following analysis hole positions and radii were randomly generated subject to non-overlap constraints as described in Section 4.2. Figure 5.3 shows $\langle F \rangle_\theta$ as a function of hole count and the overall filling fraction. For each of the fibre realisations, the filling fraction is defined as the ratio between polymer and air of the fibre cross sectional area $\frac{A_{\text{holes}}}{A_{\text{fibre}}}$. This parametrisation is by no means exhaustive, fibre geometries with similar filling fraction and number of holes can yield distinctly different scattering due to varied microstructure, variations which are visible in 5.3. Increasing both the number of holes and the filling fraction generally enhances scattering performance, driving $\langle F \rangle_\theta$

toward more negative values.

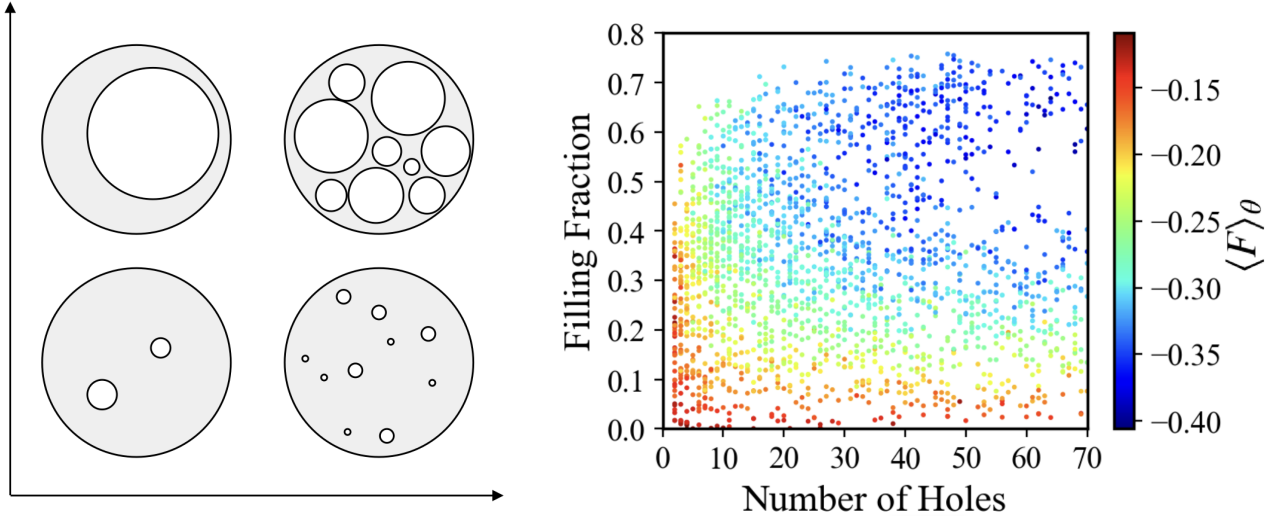


Figure 5.3: *Left:* Examples of randomly generated fibre geometries at varying filling fraction and hole count. *Right:* Distribution of randomly generated geometries ordered by filling fraction and hole count, coloured by the forward difference $\langle F \rangle_\theta$.

A minimum $\langle F \rangle_\theta = -0.40$ was identified for a structure with 64 holes and a filling fraction of 67%. We computed the Pearson correlation of the forward efficiency $\langle F \rangle_\theta$ with respect to filling fraction and number of holes finding coefficients of -0.71 and -0.86 respectively. The greater correlation with hole count suggests structural complexity plays a stronger role.

In general, structures with many air pores and a high filling fraction give superior scattering performance. However, increasing the number of pores reduces the amount of solid polymer available to bear mechanical load, making the fibre susceptible to deformation and fracture. When pores are closely spaced, the thin polymer ligaments separating them can buckle or collapse, weakening the structure and making them difficult to manufacture. For these reasons, it is desirable to identify geometries that achieve strong scattering with fewer, but more effectively positioned holes. This motivates a gradient based approach, leveraging automatic differentiation to iteratively refine pore positions and radii.

Gradient Based Optimisation

To avoid convergence to poor local minima, the optimisation was repeated from a range of random initial pore counts and configurations. Among these runs, the geometry producing the minimum $\langle F \rangle_\theta$ was retained. Since filling fraction varies during radius optimisation, each optimised geometry is compared against the best randomly generated geometry at the same hole count (Fig. 5.4).

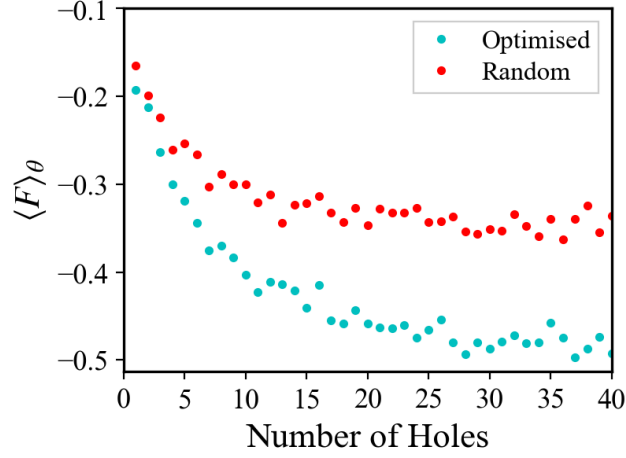


Figure 5.4: $\langle F \rangle_\theta$ versus hole count for optimised and random fibre geometries.

Optimised geometries consistently achieve stronger forward difference than randomly generated counterparts, demonstrating the efficacy of the optimisation algorithm. Random sampling begins to plateau at $\langle F \rangle_\theta = -0.3$ from 10 holes, whereas the optimised geometries approach $\langle F \rangle_\theta = -0.5$ with a plateau beyond 20 holes. As hole count increases, performance begins to saturate. This may be due to the exponentially larger parameter space that becomes increasingly difficult to explore, or because we are nearing the limit of achievable scattering in polymer porous fibres. The forward power decreases by a factor of five, from -0.11 for the solid polymer fibre to -0.50 for the optimised porous fibre. Comparable performance (~ -0.5) is achieved at a fabrication-feasible hole count of 28, where random geometries required around 70 holes to reach only -0.4 . However, this enhancement is wavelength-specific, while radiative cooling demands broadband scattering control.

5.2 Broadband Scattering

Extending the simulation framework to broadband illumination requires evaluating the scattering response over many discrete wavelengths. This extension substantially increases computational cost: the multipole matrices, translation operators, and field coefficients must be recomputed for every wavelength. Careful consideration must be given to wavelength discretization to ensure the spectral dependence of scattering is properly captured while remaining computational efficient. Radial functions arising in scattering matrices $\{J_n(k_\perp a), H_n^{(1)}(k_\perp a)\}$ oscillate with transverse wave number $k_\perp \propto k_0 = \frac{2\pi}{\lambda_0}$ which is inverse to wavelength. Therefore uniform sampling of the forward difference $\langle F \rangle_\theta$ by wave number rather than wavelength is preferred to avoid oversampling longer wavelengths. Fig. 5.5 plots the error between coarsely averaged $\langle F \rangle_{\theta,k}$ values and a reference computed with 128 discrete wave numbers.

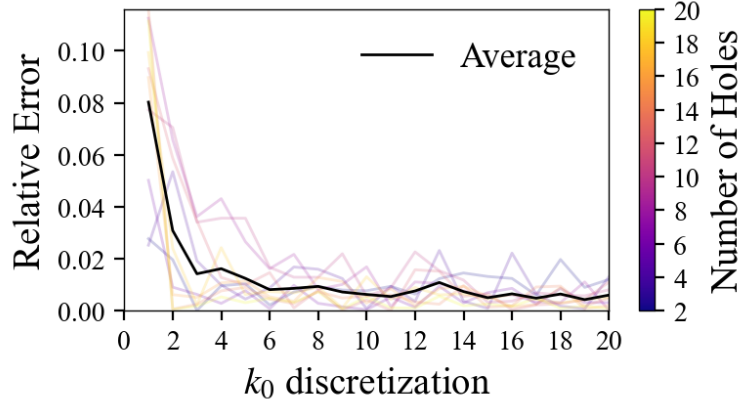


Figure 5.5: Relative error of the averaged forward difference $\langle F \rangle_{\theta,k}$ as a function of wave number k_0 discretisation. Coloured lines denote different hole counts, the black curve is the mean trend for all hole counts.

A range of geometries were tested with hole counts of 2-20. The relative error decreases sharply with increasing k_0 discretization and begins to stabilise below 1% error beyond 6 points with some fluctuation at 13. For the remainder of this work, a discretization of 15 is adopted to evaluate $\langle F \rangle_{\theta,\lambda}$ with sufficient spectral resolution while avoiding excessive computational overhead. In principle, the spectral averaging could be weighted by the solar irradiance spectrum, as certain wavelengths contribute more power from the Sun. To maintain simplicity a uniform weighting is adopted to emphasise broadband reflectance across the entire solar range rather than biasing toward regions of higher solar intensity.

5.2.1 Single Hole

Returning to a polymer fibre with a single hole, the averaged forward difference as a function of hole position x_h and radius a_h was found to be considerably less oscillatory due to the suppression of wavelength specific resonances (see Fig 5.6).

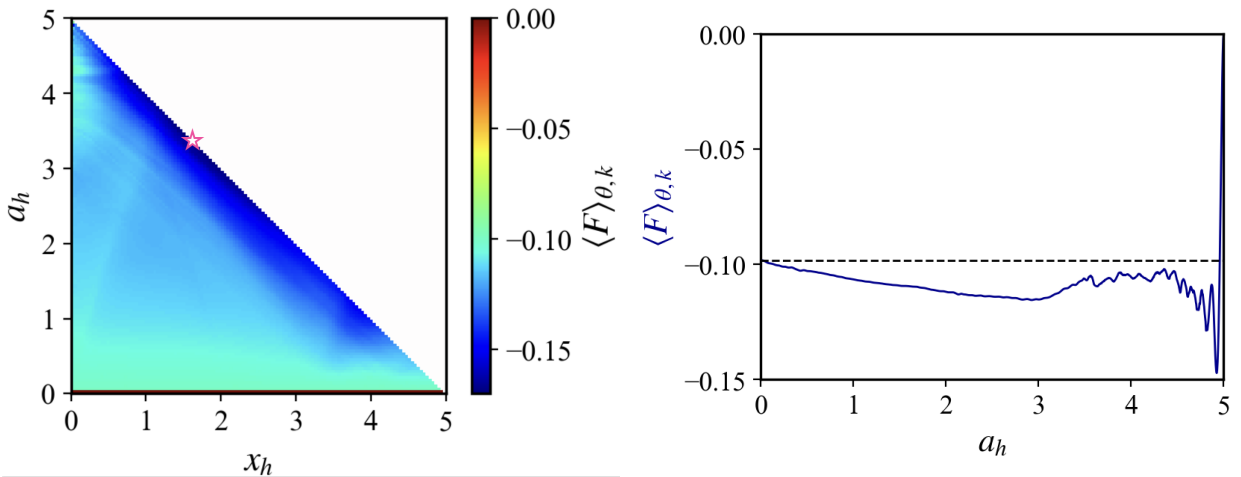


Figure 5.6: Left: $\langle F \rangle_{\theta,k}$ as a function of hole position x_h and radius a_h , minimum identified at pink star. Right: $\langle F \rangle_{\theta,k}$ as a function of hole radius a_h , centred such that $x_h = 0$.

Strong forward differences occur in regimes similar to the single wavelength case: (1) off axis holes that approach the outer boundary, and (2) large centred holes that leave only a thin polymer

shell. Optimisation isolated a minimum of $\langle F \rangle_{\theta,k} = -0.17$ for $x_h \approx 1.65\mu\text{m}$, $a_h = 3.30\mu\text{m}$ (indicated by a star in Fig. 5.6). Broadband analysis indicates symmetric, centred geometries lose their sharp resonant advantages, while, somewhat surprisingly asymmetric designs maintain smoother, wavelength independent suppression. The optimal forward difference is not as impressive as the single wavelength case ($\langle F \rangle_{\theta} = -0.22$): the need to balance competing wavelength resonances hinders the attainable scattering. Building on this, the following section extends this analysis to multi hole fibres to determine whether optimisation retains its benefit under broadband conditions.

5.2.2 Many Holes

Sampling the design space of multi hole fibres becomes significantly harder in the broadband case. A scan over a range of hole counts required roughly eight days on a standard laptop (MacBook Air, M1, 2020, 8 GB RAM). Although computationally demanding, the broadband optimisation proved worthwhile, once again yielding an improvement over random distributions.

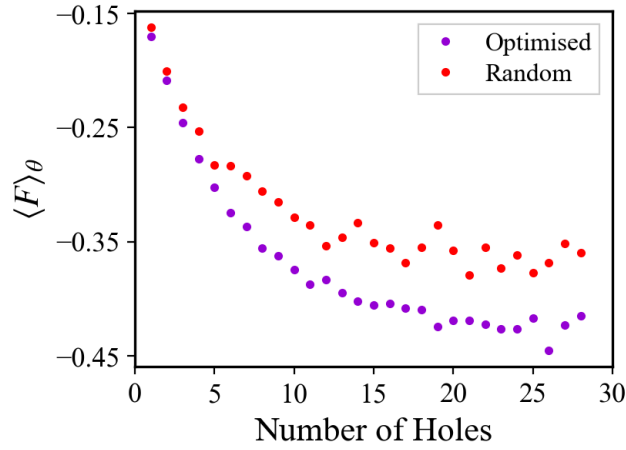


Figure 5.7: $\langle F \rangle_{\theta,k}$ at different hole counts comparing optimised and random fibre geometries.

A minimum of $\langle F \rangle_{\theta,k} = -0.45$ was achieved for a 26 hole fibre (see fibre geometry in Fig. 5.8). While the improvement over random geometries is modest, this is to be expected: random hole distributions have already been proven to produce strong solar scattering. What remains uncertain is which specific hole sizes and arrangements maximise this effect. Any additional performance gained through understanding these underlying geometric factors can have a large impact on cooling efficiency: every percentage point of bulk reflectance above 90% contributes around 10 Wm^{-2} of cooling power. To confirm that the wavenumber discretisation adequately resolved the spectral response, the best structure identified was examined over a finer discretisation of k_0 . At 128 samples, the averaged minimum shifted from $\langle F \rangle_{\theta,k} = -0.45$ to $\langle F \rangle_{\theta,k} = -0.43$, indicating coarse sampling partially overestimated the forward difference.

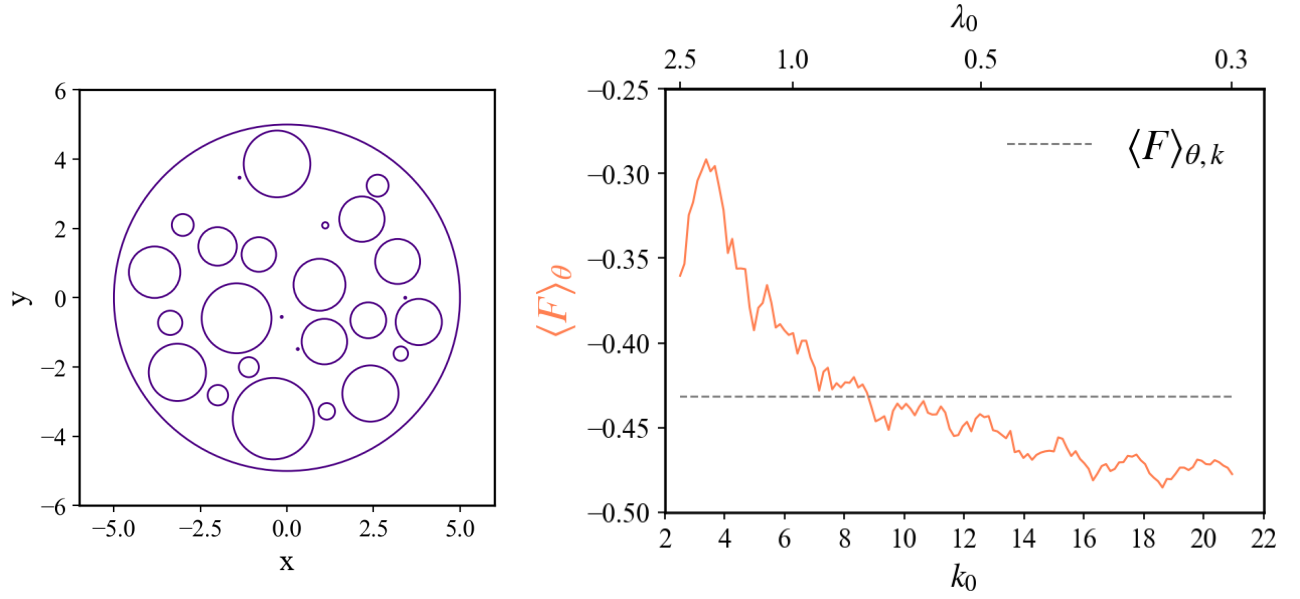


Figure 5.8: Geometry and spectral validation of the broadband optimised fibre. *Left:* best-performing 26-hole structure obtained from broadband optimisation. *Right:* Forward difference $\langle F \rangle_\theta$ plotted across wavenumber, alongside the broadband average $\langle F \rangle_{\theta,k}$ shown as a dashed line.

Examining the $\langle F \rangle_\theta$ as a function of k_0 (Fig 5.8) we see diminished scattering control at longer wavelengths (small k_0). The drop in performance may be indicative of a seeding bias: to satisfy larger hole counts, the generator favours smaller holes, which are more effective at scattering short wavelengths. Additionally, the overlap penalty indirectly limits hole growth, as neighbouring holes prevent the development of larger structures. Despite this, performance remains strong across the main solar band $\lambda_0 = 0.3\text{-}0.8 \mu\text{m}$, where the majority of solar radiance resides. The optimised geometry also exhibits a partially bipolar distribution of holes, suggesting a non random structural motif. Since multiple runs converged to similar minima with comparable performance, it is worth extending the analysis beyond a single structure. To analyse structural similarities between many optima, thirty random fibre geometries were generated using the circle packing algorithm, with the maximum hole count set to 25 to yield approximately 18 to 25 holes per structure. Each geometry was then optimised to minimise the forward scattering efficiency. Geometries that reached a forward difference below -0.4 were retained for analysis. The radii of all holes in these select fibres were extracted and combined into a single dataset which is displayed in Fig. 5.9. Optimisation often shrinks holes to negligible size when attempting to rearrange the geometry to avoid overlap, thus radii smaller than $10^{-3} \mu\text{m}$ were removed from the dataset.

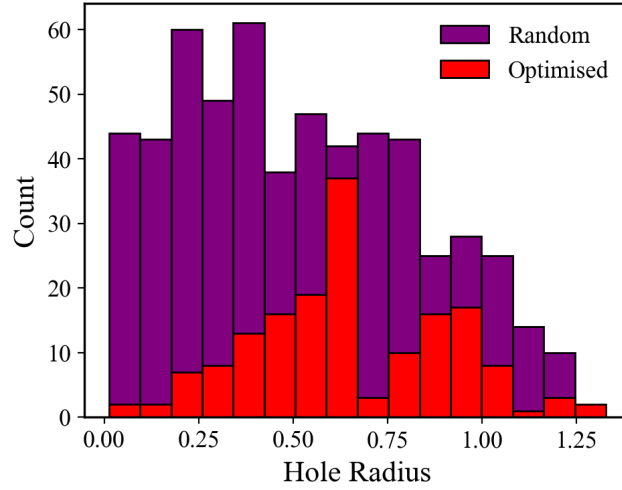


Figure 5.9: Comparison of hole radius distribution for random and optimised geometries.

The randomly generated geometries exhibit an approximately uniform distribution of pore radii at small to mid values, followed by a gradual decline toward larger radii. The drop off arises naturally from the circle-packing generator: larger holes are geometrically harder to place without overlap, so they occur less frequently as the fibre becomes crowded.

The optimised geometries show an almost bimodal distribution, with two broad peaks centred around 0.5 and 0.9 μm . The emergence of a bimodal radius distribution is consistent with previous literature on porous fibre design [58, 59] highlighting the importance of multi-scale and bimodal architectures to improve solar scattering across a range of wavelengths. The sharp 0.7 μm peak is somewhat unexpected. It could plausibly result from the discrete wavelength sampling used in optimisation, yet the spectra of individual fibres, for example Fig. 5.8, show no pronounced resonance at any wavelength. The feature therefore likely reflects a geometric scale that balances scattering performance across the solar spectrum. Comparable statistical analysis could be applied to other geometric descriptors, such as nearest neighbour distance, angular clustering, or boundary proximity to yield further insight, but this is left for future work.

6 Discussion and Future Outlook

This work set out to find the best geometry for porous radiative cooling fibres. Existing electromagnetic solvers face key limitations: FDTD and FEM can model arbitrary geometries but are computationally expensive and lack analytic gradients, RCWA offers faster evaluation but is confined to planar periodic media. Mean-field models ignore inter-fibre interactions and therefore underestimate collective scattering effects. Identifying optimal designs then requires exploring geometry precisely across every wavelength, every angle of incidence, for thousands of configurations. In this thesis, we developed a gradient aware multipole model with the efficiency required to make this exploration possible.

This was achieved by extending the multipole method, deriving new scattering matrices to be implemented in the calculation of scattered power. From the Poynting vector we realised it was possible to azimuthally average scattering over incident angle saving hundreds of simulations. We took this

averaged scattering and derived an entirely new figure of merit which describes directional scattering and therefore the cooling capability of an individual fibre. A random geometry generation protocol was developed to initialise the optimisation. We took full advantage of automatic differentiation to march random structures towards optimal cooling configurations. By comparing the many local minima we encountered, we inferred geometric motifs that promote strongly solar scattering.

Now that groundwork has been laid, exciting work is ahead. All simulations in this study were performed on a standard laptop. With greater computing power, our model could yield far more accurate and definitive conclusions. Oblique incidence averaging, larger fibre diameters, and dual purpose optimisation for infrared emission and absorption are all already implemented in the code written over this project; These have yet to be explored, and will require computing power beyond that of a laptop. Employing new random geometry seeding algorithms would allow exploration of different regions of the design space. We could extend the framework to involve many fibres interacting in a bulk material. This is a surprisingly tractable extension to the method, all fibres share the same geometry, and therefore possess identical global scattering matrices. A bulk material could therefore be modelled by rotating and translating the field basis between individual fibres. There are also clear routes to speeding up the solver itself, such as GPU acceleration, iterative scattering matrix updates, additional convolutions, and custom Bessel and Hankel function kernels.

The optimised fibre designs identified here lie within the capabilities of modern polymer drawing and extrusion, though further refinement will be needed to reach true commercial viability. Still, it's not hard to picture a world where whirring fans are replaced by clothing, tents, and tarps that silently radiate into the sky. Simulated fibres could play an elegant part in cooling the planet, one Hankel function at a time.

References

- [1] B. T. Kuhlmeier, T. P. White, G. Renversez, D. Maystre, L. C. Botten, C. M. de Sterke, and R. C. McPhedran, *J. Opt. Soc. Am. B* **19**, 2331 (2002).
- [2] G. J. M. Velders, J. S. Daniel, S. A. Montzka, I. Vimont, M. Rigby, P. B. Krummel, J. Muhle, S. O'Doherty, R. G. Prinn, R. F. Weiss, and D. Young, *Atmospheric Chemistry and Physics* **22**, 6087 (2022).
- [3] K. C. Armour, *Nature Climate Change* **7**, 331 (2017).
- [4] G. C. Simpson, *Quarterly Journal of the Royal Meteorological Society* **55**, 73 (1929).
- [5] E. Goldstein, *Radiative Sky Cooling-Enabled Efficiency Improvements on Commercial Cooling Systems*, Tech. Rep. CEC-500-2025-002 (California Energy Commission, 2024).
- [6] C. Lin, J. Hur, M. Zhang, Y. Zhang, L. Zhang, Z. Huang, X. Liu, C. Y. Li, J. Huang, F. Chu, Z. Zheng, Z. Chen, S. Yao, B. Huang, and W. Li, *Energy Conversion and Management* **315**, 118786 (2024).
- [7] M. Dong, Z. Zhang, Y. Shi, X. Zhao, S. Fan, and Z. Chen, *Nanoscale and Microscale Thermophysical Engineering* **24**, 43 (2020).
- [8] Q. Zhao, Y. Guo, T. Ye, A. Gasparrini, S. Tong, A. Overcenco, A. Urban, A. Schneider, A. Entezari, A. M. Vicedo-Cabrera, *et al.*, *The Lancet Planetary Health* **5**, e415 (2021).
- [9] B. Thompson, *Philosophical Transactions of the Royal Society of London* **1**, 139 (1800).
- [10] F. Arago, *Annuaire du Bureau des Longitudes pour l'an 1826* (Gide, 1826).
- [11] S. Catalanotti, V. Cuomo, G. Piro, D. Ruggi, V. Silvestrini, and G. Troise, *Solar Energy* **17**, 83 (1975).
- [12] A. Harrison and M. Walton, *Solar Energy* **20**, 185 (1978).
- [13] T. M. Nilsson and G. A. Niklasson, *Solar Energy Materials and Solar Cells* **37**, 93 (1995).
- [14] E. Rephaeli, A. Raman, and S. Fan, *Nano Letters* **13**, 1457 (2013).
- [15] A. P. Raman, M. A. Anoma, L. Zhu, E. Rephaeli, and S. Fan, *Nature* **515**, 540 (2014).
- [16] S. Fan and W. Li, *Nature Photonics* **16**, 182 (2022).
- [17] N. N. Shi, C.-C. Tsai, M. J. Carter, J. Mandal, A. C. Overvig, M. Y. Sfeir, M. Lu, C. L. Craig, G. D. Bernard, Y. Yang, *et al.*, *Light: Science & Applications* **7**, 37 (2018).
- [18] P.-C. Hsu, A. Y. Song, P. B. Catrysse, C. Liu, Y. Peng, J. Xie, S. Fan, and Y. Cui, *Science* **353**, 1019 (2016), <https://www.science.org/doi/pdf/10.1126/science.aaf5471>.
- [19] Y. Peng, J. Chen, A. Y. Song, P. B. Catrysse, P.-C. Hsu, L. Cai, B. Liu, Y. Zhu, G. Zhou, D. S. Wu, *et al.*, *Nature sustainability* **1**, 105 (2018).
- [20] K. Park and M. W. Frey, *Nano Letters* **23**, 10398 (2023).
- [21] S. Zeng, S. Pian, M. Su, Z. Wang, M. Wu, X. Liu, M. Chen, Y. Xiang, J. Wu, M. Zhang, Q. Cen, Y. Tang, X. Zhou, Z. Huang, R. Wang, A. Tunuhe, X. Sun, Z. Xia, M. Tian, M. Chen, X. Ma, L. Yang, J. Zhou, H. Zhou, Q. Yang, X. Li, Y. Ma, and G. Tao, *Science* **373**, 692 (2021).
- [22] B. Zhu, W. Li, Q. Zhang, D. Li, X. Liu, Y. Wang, N. Xu, Z. Wu, J. Li, X. Li, *et al.*, *Nature nanotechnology* **16**, 1342 (2021).
- [23] P. Yang, J. He, Y. Ju, Q. Zhang, Y. Wu, Z. Xia, L. Chen, and S. Tang, *Advanced Science* **10**, 2206176 (2023).
- [24] X. Zhang, W. Yang, Z. Shao, Y. Li, Y. Su, Q. Zhang, C. Hou, and H. Wang, *ACS Nano* **16**, 2188 (2022).
- [25] Y. Zhai, Y. Ma, S. N. David, D. Zhao, R. Lou, G. Tan, R. Yang, and X. Yin, *Science* **355**, 1062 (2017).
- [26] X. Wu, J. Li, F. Xie, X.-E. Wu, S. Zhao, Q. Jiang, S. Zhang, B. Wang, Y. Li, D. Gao, *et al.*,

Nature Communications **15**, 815 (2024).

- [27] D. Miao, N. Cheng, X. Wang, J. Yu, and B. Ding, *Chemical Engineering Journal* **450**, 138012 (2022).
- [28] Y. Ge, H. Jiang, K. Fu, C. Zhang, J. Zhu, C. Chen, Y. Lu, Y. Qiu, and X. Zhang, *Journal of Power Sources* **272**, 860 (2014).
- [29] A. V. Tikhonravov, M. K. Trubetskov, and G. W. DeBell, *Appl. Opt.* **35**, 5493 (1996).
- [30] P. B. Catrysse and S. Fan, *Nanophotonics* **13**, 649 (2024).
- [31] P. B. Catrysse, A. Y. Song, and S. Fan, *ACS Photonics* **3**, 2420 (2016).
- [32] T. Niu, H. Guo, J. Yu, and X. Wang, *Advanced Theory and Simulations* **7**, 2301035 (2024).
- [33] M. Li, Z. Yan, and D. Fan, *ACS Applied Materials & Interfaces* **15**, 17848 (2023).
- [34] W. Jiang, J.-Z. Liu, Z. Wang, H. Wang, X. Liu, L. Wang, X.-Q. Wang, M.-P. Zhuo, J. Hu, and K.-Q. Zhang, *Device* **3**, 100564 (2025).
- [35] M. Lei, A. Psikuta, Y. Liu, R. M. Rossi, and Y. Li, *Results in Physics* **43**, 106043 (2022).
- [36] X. Yao, L. Huang, Y. Chen, Y. Hu, K. Sagoe-Crentsil, and W. Duan, *Materials Design* **245**, 113256 (2024).
- [37] L. Wang, S. L. Jacques, and L. Zheng, *Computer Methods and Programs in Biomedicine* **47**, 131 (1995).
- [38] T. P. White, B. T. Kuhlmeier, R. C. McPhedran, D. Maystre, G. Renversez, C. M. de Sterke, and L. C. Botten, *J. Opt. Soc. Am. B* **19**, 2322 (2002).
- [39] J. W. Strutt, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* **41**, 447 (1871).
- [40] D. Felbacq, G. Tayeb, and D. Maystre, *J. Opt. Soc. Am. A* **11**, 2526 (1994).
- [41] A. W. Snyder, J. D. Love, *et al.*, *Optical waveguide theory*, Vol. 175 (Springer New York, NY, 1983).
- [42] G. N. Felder and K. M. Felder, *Mathematical Methods in Engineering and Physics*, 1st ed. (Wiley, Hoboken, NJ, 2015).
- [43] DLMF, “*NIST Digital Library of Mathematical Functions*,” Release 1.2.4 of 2025-03-15.
- [44] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. (Cambridge University Press, 2017).
- [45] M. Abramowitz, *Handbook of Mathematical Functions, With Formulas, Graphs, and Mathematical Tables*, (Dover Publications, Inc., USA, 1974).
- [46] J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang, “JAX: composable transformations of Python+NumPy programs,” (2018).
- [47] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, “Pytorch: An imperative style, high-performance deep learning library,” (2019).
- [48] M. Abadi, A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, G. S. Corrado, A. Davis, J. Dean, M. Devin, *et al.*, “Tensorflow: Large-scale machine learning on heterogeneous systems,” (2015).
- [49] Y. Saad and M. H. Schultz, *SIAM Journal on Scientific and Statistical Computing* **7**, 856 (1986).
- [50] M. F. Kasim and S. M. Vinko, “ ξ -torch: differentiable scientific computing library,” (2020).
- [51] The European Mathematical Society, “Neumann series,” (2020).
- [52] N. Wiener, *Acta mathematica* **55**, 117 (1930).
- [53] S. C. Hawkins, *ACM Trans. Math. Softw.* **46** (2020).
- [54] A. K. Gupta and S. Nadarajah, *Handbook of beta distribution and its applications* (CRC press,

2004).

- [55] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” (2019), [arXiv:1711.05101 \[cs.LG\]](#) .
- [56] C. Gueymard, D. Myers, and K. Emery, [Solar Energy](#) **73**, 443 (2002).
- [57] R. Sinclair, in *Textiles and Fashion*, Woodhead Publishing Series in Textiles, edited by R. Sinclair (Woodhead Publishing, 2015) pp. 3–27.
- [58] Q. Wang, J. Luo, Z. Lv, T. Wu, L. Zhang, Y. Zhong, H. Xu, and Z. Mao, [ACS Applied Materials & Interfaces](#) **16**, 46761 (2024).
- [59] M. Chen, D. Pang, J. Mandal, X. Chen, H. Yan, Y. He, N. Yu, and Y. Yang, [Nano Letters](#) **21**, 1412 (2021).

7 Appendix

A Scattering Matrices Derivation

At a cylindrical boundary the incoming and outgoing fields both inside and outside the boundary couple. This may be described by a set of matrices which convert between bessel and hankel coefficients. The upper index signifies the location of the field either outside the cylindrical boundary or inside the cylindrical boundary.

$$\mathbf{B}^o = R_{out}\mathbf{A}^o + T_{out}\mathbf{B}^i \quad \mathbf{A}^i = T_{in}\mathbf{A}^o + R_{in}\mathbf{B}^i \quad (\text{A.1})$$

A.1 External Matrices

To solve for the external reflection matrices we set the internal outgoing field $\mathbf{B}^i = 0$.

$$\mathbf{B}^o = R_{out}\mathbf{A}^o \quad \mathbf{A}^i = T_{in}\mathbf{A}^o \quad (\text{A.2})$$

Considering $\hat{z}$ we have

$$E_z^o - E_z^i = 0 \quad (\text{A.3})$$

$$\sum_{n \in \mathbb{Z}} \left[A_n^{Eo} J_n(k_\perp^o r) + B_n^{Eo} H_n^{(1)}(k_\perp^o r) - A_n^{Ei} J_n(k_\perp^i r) \right] e^{in\theta} e^{i(k_z z - \omega t)} = 0 \quad (\text{A.4})$$

The field is evaluated at the radius of the boundary $r = a$. The terms of the sum separate via the Fourier uniqueness theorem.

$$A_n^{Eo} J_n(k_\perp^o a) + B_n^{Eo} H_n^{(1)}(k_\perp^o a) - A_n^{Ei} J_n(k_\perp^i a) = 0 \quad (\text{A.5})$$

And by a similar derivation

$$A_n^{Ko} J_n(k_\perp^o a) + B_n^{Ko} H_n^{(1)}(k_\perp^o a) - A_n^{Ki} J_n(k_\perp^i a) = 0 \quad (\text{A.6})$$

Returning to the Boundary Conditions we now consider $\hat{\theta}$, and substitute our equations for E_θ, K_θ in terms of E_z, K_z .

$$E_\theta^o = E_\theta^i \quad * \quad (\text{A.7})$$

$$K_\theta^o = K_\theta^i \quad ** \quad (\text{A.8})$$

$$\frac{i}{k_\perp^{o2}} \left[\frac{k_z}{a} \frac{\partial}{\partial \theta} E_z^o - k_0 \mu_o \frac{\partial}{\partial r} K_z^o \right] = \frac{i}{k_\perp^{i2}} \left[\frac{k_z}{a} \frac{\partial}{\partial \theta} E_z^i - k_0 \mu_i \frac{\partial}{\partial r} K_z^i \right] \quad * \quad (\text{A.9})$$

$$\frac{i}{k_\perp^{o2}} \left[\frac{k_z}{a} \frac{\partial}{\partial \theta} K_z^o + k_0 \epsilon_o \frac{\partial}{\partial r} E_z^o \right] = \frac{i}{k_\perp^{i2}} \left[\frac{k_z}{a} \frac{\partial}{\partial \theta} K_z^i + k_0 \epsilon_i \frac{\partial}{\partial r} E_z^i \right] \quad ** \quad (\text{A.10})$$

Rearranging

$$\frac{k_z}{a} \frac{\partial}{\partial \theta} (k_\perp^{i2} E_z^o - k_\perp^{o2} E_z^i) - k_0 \frac{\partial}{\partial r} (k_\perp^{i2} \mu_o K_z^o - k_\perp^{o2} \mu_i K_z^i) = 0 \quad * \quad (\text{A.11})$$

$$\frac{k_z}{a} \frac{\partial}{\partial \theta} (k_{\perp}^i{}^2 K_z^o - k_{\perp}^o{}^2 K_z^i) + k_0 \frac{\partial}{\partial r} (k_{\perp}^i{}^2 \epsilon_o E_z^o - k_{\perp}^o{}^2 \epsilon_i E_z^i) = 0 \quad * * \quad (\text{A.12})$$

Substituting the E_z fields (A.4)

$$\sum_{n \in \mathbb{Z}} \frac{k_z}{a} \frac{\partial}{\partial \theta} \left[\left(k_{\perp}^i{}^2 A_n^{Eo} J_n(k_{\perp}^o r) + k_{\perp}^i{}^2 B_n^{Eo} H_n^{(1)}(k_{\perp}^o r) - k_{\perp}^o{}^2 A_n^{Ei} J_n(k_{\perp}^i r) \right) e^{in\theta} \right] - k_0 \frac{\partial}{\partial r} \left[\left(k_{\perp}^i{}^2 \mu_o A_n^{Ko} J_n(k_{\perp}^o r) + k_{\perp}^i{}^2 \mu_o B_n^{Ko} H_n^{(1)}(k_{\perp}^o r) - k_{\perp}^o{}^2 \mu_i A_n^{Ki} J_n(k_{\perp}^i r) \right) e^{in\theta} \right] = 0 \quad * \quad (\text{A.13})$$

In the following I employ $J^i = J_n(k_{\perp}^i a)$, $J^o = J_n(k_{\perp}^o a)$, $H^o = H_n^{(1)}(k_{\perp}^o a)$.

$$\frac{k_z in}{a} \left[k_{\perp}^i{}^2 A^{Eo} J^o + k_{\perp}^i{}^2 B^{Eo} H^o - k_{\perp}^o{}^2 A^{Ei} J^i \right] - k_0 \left[k_{\perp}^i{}^2 k_{\perp}^o \mu_o A^{Ko} J'^o + k_{\perp}^i{}^2 k_{\perp}^o \mu_o B^{Ko} H'^o - k_{\perp}^o{}^2 k_{\perp}^i \mu_i A^{Ki} J'^i \right] = 0 \quad * \quad (\text{A.14})$$

Substituting the z boundary conditions.

$$\frac{k_z in}{ak_0} (k_{\perp}^i{}^2 - k_{\perp}^o{}^2) A^{Ei} J^i - \left[k_{\perp}^i{}^2 k_{\perp}^o \mu_o A^{Ko} J'^o + k_{\perp}^i{}^2 k_{\perp}^o \mu_o B^{Ko} H'^o - k_{\perp}^o{}^2 k_{\perp}^i \mu_i A^{Ki} J'^i \right] = 0 \quad * \quad (\text{A.15})$$

I define the following.

$$C = \frac{k_z in (k_{\perp}^i{}^2 - k_{\perp}^o{}^2)}{ak_0 k_{\perp}^i k_{\perp}^o} \quad (\text{A.16})$$

Thus

$$C A^{Ei} J^i - k_{\perp}^i \mu_o (A^{Ko} J'^o + B^{Ko} H'^o) + k_{\perp}^o \mu_i A^{Ki} J'^i = 0 \quad * \quad (\text{A.17})$$

and by similar manipulation

$$-C A^{Ki} J^i - k_{\perp}^i \epsilon_o (A^{Eo} J'^o + B^{Eo} H'^o) + k_{\perp}^o \epsilon_i A^{Ei} J'^i = 0 \quad * * \quad (\text{A.18})$$

External Transmission

First we solve for the transmission matrix substituting B terms using the E_z boundary condition.

$$\mathbf{A}^i = T_{in} \mathbf{A}^o \quad (\text{A.19})$$

$$B^{Eo} = \frac{A^{Ei} J^i - A^{Eo} J^o}{H^{Eo}} \quad B^{Ko} = \frac{A^{Ki} J^i - A^{Ko} J^o}{H^{Ko}} \quad (\text{A.20})$$

The substitution of which yields

$$A^{Ei} = \frac{k_{\perp}^i \mu_o A^{Ko} W^o + X_{\mu}^{JH} A^{Ki}}{C J^i H^o} \quad * \quad (\text{A.21})$$

$$A^{Ki} = \frac{k_{\perp}^i \epsilon_o A^{Eo} W^o + X_{\epsilon}^{JH} A^{Ei}}{-C J^i H^o} \quad * * \quad (\text{A.22})$$

Where W^o is the [wronksian](#)

$$W\{J^o, H^o\} = J'^o H^o - H'^o J^o = \frac{2i}{\pi k_{\perp}^o a} \quad (\text{A.23})$$

I also define the following cross-terms.

$$X_{\epsilon}^{JH} = k_{\perp}^o \epsilon_i J^i H^o - k_{\perp}^i \epsilon_o J^i H'^o \quad X_{\mu}^{HH} = k_{\perp}^o \mu_i H^i H^o - k_{\perp}^i \mu_o H^i H'^o \quad (\text{A.24})$$

Each reflection/transmission matrix has the same denominator.

$$\delta = (CJ^i H^o)^2 + X_{\epsilon}^{JH} X_{\mu}^{JH} \quad (\text{A.25})$$

The transmission matrix is comprised of four elements which describe the cross coupling between the electric and magnetic field.

$$\begin{bmatrix} A^{Ei} \\ A^{Ki} \end{bmatrix} = \begin{bmatrix} T_{in}^{EE} & T_{in}^{EK} \\ T_{in}^{KE} & T_{in}^{KK} \end{bmatrix} \begin{bmatrix} A^{Eo} \\ A^{Ko} \end{bmatrix} \quad (\text{A.26})$$

$$T_{in}^{EE} = \frac{-k_{\perp}^i \epsilon_o W^o X_{\mu}^{JH}}{\delta} \quad (\text{A.27})$$

$$T_{in}^{EK} = \frac{k_{\perp}^i \mu_o W^o C J^i H^o}{\delta} \quad (\text{A.28})$$

$$T_{in}^{KE} = \frac{k_{\perp}^i \epsilon_o W^o C J^i H^o}{\delta} \quad (\text{A.29})$$

$$T_{in}^{KK} = \frac{-k_{\perp}^i \mu_o W^o X_{\epsilon}^{JH}}{\delta} \quad (\text{A.30})$$

External Reflection

In a similar fashion reflection matrices may be garnered by initially eliminating A^{Ei} , A^{Ki} and solving for B^{Eo} , B^{Ko} instead.

$$B^{Eo} = \frac{CJ^o J^i A^{Eo} + X_{\mu}^{JH} B^{Ko} + X_{\mu}^{JJ} A^{Ko}}{-CH^o J^i} \quad (\text{A.31})$$

$$B^{Ko} = \frac{-CJ^o J^i A^{Ko} + X_{\epsilon}^{JH} B^{Eo} + X_{\epsilon}^{JJ} A^{Eo}}{CH^o J^i} \quad (\text{A.32})$$

Which may be substituted to yield.

$$R_{out}^{EE} = \frac{-C^2 H^o J^o J^{i2} - X_{\mu}^{JH} X_{\epsilon}^{JJ}}{\delta} \quad (\text{A.33})$$

$$R_{out}^{KK} = \frac{-C^2 H^o J^o J^{i2} - X_{\epsilon}^{JH} X_{\mu}^{JJ}}{\delta} \quad (\text{A.34})$$

$$R_{out}^{EK} = \frac{-k_{\perp}^i \mu_o W^o C J^{i2}}{\delta} \quad (\text{A.35})$$

$$R_{out}^{KE} = \frac{k_{\perp}^i \epsilon_o W^o C J^i{}^2}{\delta} \quad (\text{A.36})$$

A.2 Internal Matrices

Returning to the very beginning I instead set the field outside the cylinder $\mathbf{A}^o = 0$ to solve for the internal matrices.

$$\mathbf{B}^o = T_{out} \mathbf{B}^i \quad \mathbf{A}^i = R_{in} \mathbf{B}^i \quad (\text{A.37})$$

The theta boundary condition for the electric field now looks like

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{k_z}{a} \frac{\partial}{\partial \theta} \left[\left(k_{\perp}^i{}^2 B_n^{Eo} H_n^{(1)}(k_{\perp}^o r) - k_{\perp}^o{}^2 A_n^{Ei} J_n(k_{\perp}^i r) - k_{\perp}^o{}^2 B_n^{Ei} H_n^{(1)}(k_{\perp}^i r) \right) e^{in\theta} \right] \\ - k_0 \frac{\partial}{\partial r} \left[\left(k_{\perp}^i{}^2 \mu_o B_n^{Ko} H_n^{(1)}(k_{\perp}^o r) - k_{\perp}^o{}^2 \mu_i A_n^{Ki} J_n(k_{\perp}^i r) - k_{\perp}^o{}^2 \mu_i B_n^{Ki} H_n^{(1)}(k_{\perp}^i r) \right) e^{in\theta} \right] = 0 \quad * \end{aligned} \quad (\text{A.38})$$

Yielding two similar equations to the external matrices where the i/o are flipped and therefore the sign of C changes, the outer AJ terms also become B terms.

$$-CB^{Eo}H^o - k_{\perp}^o \mu_i (A^{Ki}J^i + B^{Ki}H^i) + k_{\perp}^i \mu_o B^{Ko}H'^o = 0 \quad * \quad (\text{A.39})$$

$$CB^{Ko}H^o - k_{\perp}^o \epsilon_i (A^{Ei}J^i + B^{Ei}H^i) + k_{\perp}^i \epsilon_o B^{Eo}H'^o = 0 \quad ** \quad (\text{A.40})$$

Internal Transmission

$$\mathbf{B}^o = T_{out} \mathbf{B}^i \quad (\text{A.41})$$

Substitute out A^i terms and solve for B^o

$$B^{Eo} = \frac{k_{\perp}^o \mu_i B^{Ki} W^i + X_{\mu} A^{Ko}}{C J^i H^o} \quad * \quad (\text{A.42})$$

$$A^{Ko} = \frac{k_{\perp}^o \epsilon_i A^{Ei} W^i + X_{\epsilon} A^{Eo}}{-C J^i H^o} \quad ** \quad (\text{A.43})$$

Thus the internal transmission matrix elements are

$$T_{out}^{EE} = \frac{-k_{\perp}^o \epsilon_i W^i X_{\mu}^{JH}}{\delta} \quad (\text{A.44})$$

$$T_{out}^{KK} = \frac{-k_{\perp}^o \mu_i W^i X_{\epsilon}^{JH}}{\delta} \quad (\text{A.45})$$

$$T_{out}^{EK} = \frac{k_{\perp}^o \mu_i W^i C J^i H^o}{\delta} \quad (\text{A.46})$$

$$T_{out}^{KE} = \frac{-k_{\perp}^o \epsilon_i W^i C J^i H^o}{\delta} \quad (\text{A.47})$$

Internal Reflection

Solving for B^i by removing B^o terms results in the following coupled equations

$$A^{Ei} = \frac{-CH^i H^o B^{Ei} + X_\mu^{JH} A^{Ki} + X_\mu^{HH} B^{Ki}}{CJ^i H^o} \quad (\text{A.48})$$

$$A^{Ki} = \frac{CH^i H^o B^{Ki} + X_\epsilon^{JH} A^{Ei} + X_\epsilon^{HH} B^{Ei}}{-CJ^i H^o} \quad (\text{A.49})$$

Therefore the internal reflection matrices are as follows:

$$R_{in}^{EE} = \frac{-C^2 H^i J^i H^{o2} - X_\mu^{JH} X_\epsilon^{HH}}{\delta} \quad (\text{A.50})$$

$$R_{in}^{KK} = \frac{-C^2 H^i J^i H^{o2} - X_\epsilon^{JH} X_\mu^{HH}}{\delta} \quad (\text{A.51})$$

$$R_{in}^{EK} = \frac{-k_\perp^o \mu_i C W^i H^{o2}}{\delta} \quad (\text{A.52})$$

$$R_{in}^{KE} = \frac{k_\perp^o \epsilon_i C W^i H^{o2}}{\delta} \quad (\text{A.53})$$

B Scattering Power Derivation

$$\langle S_r \rangle = \frac{-k_0}{2Z_0 k_\perp^2} \Im \left(E_z \frac{\partial E_z^*}{\partial r} + K_z \frac{\partial K_z^*}{\partial r} \right) \quad (\text{B.1})$$

Taking a single term

$$\Im \left(E_z \frac{\partial}{\partial r} E_z^* \right) = \Im \left((E_z^i + E_z^s) \frac{\partial}{\partial r} (E_z^i + E_z^s)^* \right) \quad (\text{B.2})$$

The following is equivalent for the E or K field, the radial derivative imparts a $k_\perp$ via the chain rule.

$$= \Im \left((A_n J_n + B_n H_n) e^{in\theta} \frac{\partial}{\partial r} (A_m J_m + B_m H_m)^* e^{-im\theta} \right) \quad (\text{B.3})$$

$$= k_\perp \cdot \Im \left((A_n A_m^* J_n J_m'^* + A_n B_m^* J_n H_m'^* + B_n A_m^* H_n J_m'^* + B_n B_m^* H_n H_m'^*) e^{i(n-m)\theta} \right) \quad (\text{B.4})$$

The leading order [asymptotic expansion](#) of the Hankel function (of the first kind) may be used to simplify derivatives in the far field approximation.

$$H_n(u) \simeq \sqrt{\frac{2}{\pi u}} e^{i(u - \frac{\pi}{2}n - \frac{\pi}{4})} \quad u \gg 1 \quad (\text{B.5})$$

$$H'_n(u) \simeq i H_n^{(1)}(u) \quad u \gg 1 \quad (\text{B.6})$$

$$J_n = \frac{1}{2} (H_n + H_n^*) \quad (\text{B.7})$$

$$J'_n = \frac{i}{2}(H_n - H_n^*) \quad (\text{B.8})$$

A further simplification is possible if we consider a $\tilde{k}_\perp = k_\perp + \eta i$

$$H_n(k_\perp^+ r) H_m(k_\perp^+ r) \propto e^{2i\tilde{k}_\perp r} = e^{2ik_\perp r} e^{-2\eta r} \quad (\text{B.9})$$

In the limit as $r \rightarrow \infty$ these terms go to zero, this may be followed by the limit of $\eta \rightarrow 0$ to retain the lossless convention.

$$k_\perp \Im((A_n A_m^* J_n J_m'^* + A_n B_m^* J_n H_m'^* + B_n A_m^* H_n J_m'^* + B_n B_m^* H_n H_m'^*) e^{i(n-m)\theta}) \quad (\text{B.10})$$

$$= k_\perp \Im \left((A_n A_m^* \frac{i}{4} (H_n + H_n^*) (H_m - H_m^*) + A_n B_m^* \frac{1}{2} (H_n + H_n^*) (-i) H_m^* + B_n A_m^* H_n \frac{-i}{2} (H_m^* - H_m) + B_n B_m^* H_n (-i) H_m^*) e^{i(n-m)\theta} \right) \quad (\text{B.11})$$

$$= k_\perp \Im \left(\left(\frac{1}{4} A_n A_m^* (H_n^* H_m - H_n H_m^*) - \frac{1}{2} A_n B_m^* H_n H_m^* - \frac{1}{2} B_n A_m^* H_n H_m^* - B_n B_m^* H_n H_m^* \right) e^{i(n-m)\theta} \right) \quad (\text{B.12})$$

$$\Im(i(a + bi)) = \Re(a + bi) \quad (\text{B.13})$$

$$= k_\perp \Re \left(\frac{1}{4} A_n A_m^* (H_n^* H_m - H_n H_m^*) e^{i(n-m)\theta} - (\Re(A_n B_m^*) + B_n B_m^*) H_n H_m^* e^{i(n-m)\theta} \right) \quad (\text{B.14})$$

$$H_n H_m^* \simeq \frac{2}{\pi u} e^{i(n-m)(-\frac{\pi}{2})} \quad (\text{B.15})$$

$$= k_\perp \cdot \frac{2}{\pi k_\perp r} \Re \left(\frac{1}{4} A_n A_m^* (e^{i(n-m)(\frac{\pi}{2})} - e^{i(n-m)(-\frac{\pi}{2})}) e^{i(n-m)\theta} - (\Re(A_n B_m^*) + B_n B_m^*) e^{i(n-m)(\theta-\frac{\pi}{2})} \right) \quad (\text{B.16})$$

$$= k_\perp \cdot \frac{2}{\pi k_\perp r} \Re \left(\frac{1}{4} A_n A_m^* (e^{i(n-m)(\pi)} - e^{i(n-m)(0)}) e^{i(n-m)(\theta-\frac{\pi}{2})} - (\Re(A_n B_m^*) + B_n B_m^*) e^{i(n-m)(\theta-\frac{\pi}{2})} \right) \quad (\text{B.17})$$

$$= \frac{2}{\pi r} \Re \left(\left(\frac{1}{4} A_n A_m^* [(-1)^{n-m} - 1] - \Re(A_n B_m^*) - B_n B_m^* \right) e^{i(n-m)(\theta-\frac{\pi}{2})} \right) \quad (\text{B.18})$$

The total poynting vector is then,

$$\langle S_r \rangle = \frac{-k_0}{Z_0 k_\perp^2 \pi r} \Re \left(\left(\frac{1}{4} (A_n^E A_m^{E*} + A_n^K A_m^{K*}) [(-1)^{n-m} - 1] - \Re(A_n^E B_m^{E*} + A_n^K B_m^{K*}) - B_n^E B_m^{E*} - B_n^K B_m^{K*} \right) e^{i(n-m)(\theta-\frac{\pi}{2})} \right) \quad (\text{B.19})$$

B.1 Forward-Half Scattering Cross-Section

At this stage an integral over the half plane either side of the incident plane wave may be taken to determine the total power radiated in the forward direction.

$$\int_{\theta_k - \frac{\pi}{2}}^{\theta_k + \frac{\pi}{2}} e^{i(n-m)\theta} r d\theta \quad (\text{B.20})$$

$$= e^{i(n-m)\theta_k} \frac{e^{i(n-m)(\frac{\pi}{2})} - e^{i(n-m)(-\frac{\pi}{2})}}{i(n-m)} \quad (\text{B.21})$$

$$= \pi r e^{i(n-m)\theta_k} \text{sinc}\left((n-m)\frac{\pi}{2}\right) \quad (\text{B.22})$$

This yields the power radiated in the forward direction per unit length, where $\text{sinc } x = \frac{\sin x}{x}$

$$P_f = \frac{-k_0}{Z_0 k_{\perp}^2} \Re \left(\left(\frac{1}{4} (A_n^E A_m^{E*} + A_n^K A_m^{K*}) [(-1)^{n-m} - 1] \right. \right. \\ \left. \left. - \Re(A_n^E B_m^{E*} + A_n^K B_m^{K*}) - B_n^E B_m^{E*} - B_n^K B_m^{K*} \right) e^{i(n-m)(\theta_k - \frac{\pi}{2})} \text{sinc}\left((n-m)\frac{\pi}{2}\right) \right) \quad (\text{B.23})$$

For simplicity I consider averaging a single term over all incident angles,

$$\frac{1}{2\pi} \int_0^{2\pi} A_n^E B_m^{E*} e^{i(n-m)\theta_k} d\theta_k \quad (\text{B.24})$$

$$= \frac{1}{2\pi} V_{\perp}^2 \int_0^{2\pi} \sin(\delta) e^{in(\frac{\pi}{2} - \theta_k)} (S_{m,p}^{*EE} \sin(\delta) + S_{m,p}^{*EK} \cos(\delta)) e^{-ip(\frac{\pi}{2} - \theta_k)} e^{i(n-m)\theta_k} d\theta_k \quad (\text{B.25})$$

The transverse amplitude is defined $V_{\perp} = V_0 \sin(\phi_k)$ The parameter δ describes the polarization of the incident field, which may also be averaged.

$$\frac{1}{2\pi} \int_0^{2\pi} \sin^2(\delta) d\delta = \frac{1}{2\pi} \int_0^{2\pi} \cos^2(\delta) d\delta = \frac{1}{2} \quad \frac{1}{2\pi} \int_0^{2\pi} \sin(\delta) \cos(\delta) d\delta = 0 \quad (\text{B.26})$$

$$= \frac{1}{4\pi} V_{\perp}^2 \int_0^{2\pi} e^{i(n-p)\frac{\pi}{2}} S_{m,p}^{*EE} e^{i(p-m)\theta_k} d\theta_k \quad (\text{B.27})$$

$$= \frac{1}{4\pi} V_{\perp}^2 e^{i(n-p)\frac{\pi}{2}} S_{m,p}^{*EE} 2\pi \delta_{m,p} \quad (\text{B.28})$$

$$= \frac{V_{\perp}^2}{2} S_{m,m}^{*EE} e^{i(n-m)\frac{\pi}{2}} \quad (\text{B.29})$$

We can also consider the B-B terms

$$\frac{1}{2\pi} \int_0^{2\pi} B_n^E B_m^{E*} e^{i(n-m)\theta_k} d\theta_k \quad (\text{B.30})$$

$$= \frac{1}{2\pi} V_{\perp}^2 \int_0^{2\pi} e^{iq(\frac{\pi}{2} - \theta_k)} (S_{n,q}^{*EE} \sin(\delta) + S_{n,q}^{*EK} \cos(\delta)) (S_{m,p}^{*EE} \sin(\delta) + S_{m,p}^{*EK} \cos(\delta)) e^{-ip(\frac{\pi}{2} - \theta_k)} e^{i(n-m)\theta_k} d\theta_k \quad (\text{B.31})$$

$$= \frac{V_{\perp}^2}{2} (S_{n,q}^{EE} S_{m,p}^{*EE} + S_{n,q}^{EK} S_{m,p}^{*EK}) e^{i(n-m)\frac{\pi}{2}} \delta_{n-m,q-p} \quad (\text{B.32})$$

and finally A-A terms

$$\frac{1}{2\pi} \int_0^{2\pi} A_n^E A_m^{E*} e^{i(n-m)\theta_k} d\theta_k \quad (\text{B.33})$$

$$= \frac{1}{2\pi} V_{\perp}^2 \int_0^{2\pi} e^{in(\frac{\pi}{2}-\theta_k)} \sin^2(\delta) e^{-im(\frac{\pi}{2}-\theta_k)} e^{i(n-m)\theta_k} d\theta_k \quad (\text{B.34})$$

$$= \frac{V_{\perp}^2}{4\pi} \int_0^{2\pi} e^{i(n-m)(\frac{\pi}{2})} d\theta_k \quad (\text{B.35})$$

$$= \frac{V_{\perp}^2}{2} e^{i(n-m)(\frac{\pi}{2})} \quad (\text{B.36})$$

The power scattered forward averaged over any incident plane-wave is

$$\begin{aligned} \langle P_f \rangle_{\theta_k} = & \frac{-k_0 V_{\perp}^2}{2Z_0 k_{\perp}^2} \Re \left(\frac{(-1)^{n-m} - 1}{4} \text{sinc} \left((n-m) \frac{\pi}{2} \right) - 2\Re(S_{m,m}^{EE} + S_{m,m}^{KK}) \right. \\ & \left. - (S_{n,q}^{EE} S_{m,p}^{*EE} + S_{n,q}^{EK} S_{m,p}^{*EK} + S_{n,q}^{KE} S_{m,p}^{*KE} + S_{n,q}^{KK} S_{m,p}^{*KK}) \delta_{n-m,q-p} \right] \text{sinc} \left((n-m) \frac{\pi}{2} \right) \end{aligned} \quad (\text{B.37})$$

The factor of 2 for the cross term comes from the following identity.

$$\sum_d \text{sinc}(d \frac{\pi}{2}) = 2 \quad (\text{B.38})$$

alongside

$$\frac{(-1)^{n-m} - 1}{4} \text{sinc} \left((n-m) \frac{\pi}{2} \right) = -\frac{1}{2} \text{sinc} \left((n-m) \frac{\pi}{2} \right) \quad (\text{B.39})$$

Simplifying the equation to

$$\begin{aligned} \langle P_f \rangle_{\theta_k} = & \frac{-k_0 V_{\perp}^2}{2Z_0 k_{\perp}^2} \Re \left(-\frac{1}{2} \text{sinc} \left((n-m) \frac{\pi}{2} \right) - 2\Re(S_{m,m}^{EE} + S_{m,m}^{KK}) \right. \\ & \left. - (S_{n,q}^{EE} S_{m,p}^{*EE} + S_{n,q}^{EK} S_{m,p}^{*EK} + S_{n,q}^{KE} S_{m,p}^{*KE} + S_{n,q}^{KK} S_{m,p}^{*KK}) \delta_{n-m,q-p} \text{sinc} \left((n-m) \frac{\pi}{2} \right) \right) \end{aligned} \quad (\text{B.40})$$

The plane wave/source Term is divergent, the integral over a half plane of infinite radius yields infinite incident power, for the figure of merit we consider the difference in average forward suppression power:

$$\begin{aligned} \langle \tilde{P}_f \rangle_{\theta_k} = & \frac{k_0 V_{\perp}^2}{2Z_0 k_{\perp}^2} \Re [2(S_{m,m}^{EE} + S_{m,m}^{KK}) \\ & + (S_{n,q}^{EE} S_{m,p}^{*EE} + S_{n,q}^{EK} S_{m,p}^{*EK} + S_{n,q}^{KE} S_{m,p}^{*KE} + S_{n,q}^{KK} S_{m,p}^{*KK}) \delta_{n-m,q-p} \text{sinc} \left((n-m) \frac{\pi}{2} \right)] \end{aligned} \quad (\text{B.41})$$

The intensity of a plane-wave is given by

$$I_0 = \frac{E_0^2}{2Z_0} = \frac{V_{\perp}^2 k_0}{2Z_0 k_{\perp}} \quad (\text{B.42})$$

Where $V_{\perp} = E_0 \sin(\phi_{inc})$ is the 2D projection of the field strength. The forward cross section and quality are given by.

$$\langle \sigma \rangle_{\theta} = \frac{\langle \tilde{P}_f \rangle_{\theta}}{I_0} \quad \langle F \rangle_{\theta} = \frac{\langle \sigma \rangle_{\theta}}{2a} \quad (\text{B.43})$$

Where a is the radius of the scattering fiber/structure. Thus the figure of merit is the averaged forward suppression quality, a dimensionless quality which measures the amount of light removed from the source plane-wave in its direction of incidence.

$$\langle F \rangle_{\theta} = \frac{1}{2k_{\perp}a} \Re [2(S_{m,m}^{EE} + S_{m,m}^{KK}) + (S_{n,q}^{EE} S_{m,p}^{*EE} + S_{n,q}^{EK} S_{m,p}^{*EK} + S_{n,q}^{KE} S_{m,p}^{*KE} + S_{n,q}^{KK} S_{m,p}^{*KK}) \delta_{n-m,q-p} \text{sinc}((n-m)\frac{\pi}{2})] \quad (\text{B.44})$$

C Computational Aspects

The solver and optimiser developed for this project have been compiled into a public GitHub repository (available at github.com/danielglass1/Daniel-Glass-Honours). The folder, `mp_torch`, contains the components of the multipole solver and the file `optimiser.py` is the optimisation routine. An example script using the solver to generate a field plot of a porous fibre is provided below.

```
from mp_torch.generate_field import generate_field
from mp_torch.solver import solver
from mp_torch.forward import forward
import torch
import matplotlib.pyplot as plt
import matplotlib.patches as patches

import matplotlib
matplotlib.use('MacOSX') # or 'Qt5Agg', 'MacOSX', etc.

### INCIDENT FIELD VAR #####
inc_phi = torch.tensor(0.5*torch.pi) # oblique angle
inc_theta = torch.tensor(0*torch.pi) # azimuthal angle
inc_lam=1 # wavelength
inc_k_0 = torch.tensor(2*torch.pi)/inc_lam # wavenumber
inc_k_perp=inc_k_0*torch.sin(inc_phi) # perpendicular wavenumber
inc_magnitude=1 # field amplitude (V_0)
inc_delta=torch.tensor(0.25*torch.pi) # polarization angle

### STRUCTURE #####
# fibre # n - fibre refractive index
fibre_n_real = 1.5 # Re(n)
```



```

fibre_n_imag= 0                                # Im(n)
fibre_a = torch.tensor(5)                      # fibre radius

# holes
#
#           X   Y   R   Re(n)   Im(n)
holes=torch.tensor([
    [ -1,  -3,  1,    1,    0],
    [ -3,  1,  1.2,   1,    0],
    [  2,  2,  2,    1,    0],
],dtype=torch.float32)

### GENERATE FIELD #####
bessel_trunc=1.0

#holes, wavenumber, phi, fibre radius, fibre Re(n), fibre Im(n), bessel_truncation)
solver1=solver(holes, inc_k_0, inc_phi, fibre_a, fibre_n_real, \
fibre_n_imag, bessel_trunc)
S_0=solver1.jacket_B_matrix

avg_F_theta=forward(S_0,inc_k_perp,fibre_a) #Figure of Merit - Forward Difference
print(avg_F_theta)

### PLOTTING #####

npts=128
xy_range=fibre_a*2
E_field, K_field = generate_field(solver1,inc_magnitude,inc_delta, \
inc_theta,xy_range,npts)

plot_extent = [-xy_range, xy_range, -xy_range, xy_range]

# Plot Electric Field
im=plt.imshow(torch.real(E_field), extent=plot_extent, origin='lower', \
cmap='vanimo', vmax=2,vmin=-2)
plt.colorbar()
plt.xlabel("x")
plt.ylabel("y")
plt.gca().add_patch(patches.Circle((0, 0), fibre_a, edgecolor='white', \
facecolor='none', linewidth=0.2))

```

```
for x,y,r,_,_ in holes:
    plt.gca().add_patch(patches.Circle((x, y), r, edgecolor='white', \
        facecolor='none', linewidth=0.2))
plt.show()

# Plot Magnetic Field
# im.set_data(torch.real(K_field))
# plt.show()
```